Kaynaklı Birleşimler
Weld Connection

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Birleşim elemanları (Connection Methods)

- Perçin





- Bulon (cıvata)



- Kaynak






000000 29/09/2016 0.379953 s 5000 fps 5 µs 0.000 1016x916





Kaynaklı birleşimler (Weld Connection)

- Kaynak, malzemelerin ısı ve/veya basınç uygulanarak birleştirilmesine denir.
- Bazı kaynak yöntemlerinde kaynak elektrotu, kaynak teli gibi ilaveler kullanılır.
- Kaynak işlemi koruyucu gaz, kaynak tozu yada pasta gibi yardımcı malzemeler ile kolaylaştırılabilir. Kaynak için gerekli enerji birleşime dışarıdan uygulanır.
- Kaynaklı birleşimler hasar vermeden çözülemeyen tip birleşimlerdir.

- Welding is the combination of materials by applying heat and / or pressure.
- In some welding methods, additions such as welding electrode and welding wire are used.
- Welding process can be facilitated by auxiliary materials such as shielding gas, flux or paste. The energy required for welding is applied externally to the joint.
- Welded joints are type joints that cannot be dissolved without damage.

Kaynak Türleri (Types of Weld)

- Eritme Kaynağı: ısı derecesinin, metallerin **ergime** noktasına kadar (metallerin sıvı kıvama getirecek kadar) yükseltilmesiyle gerçekleşen kaynaktır. (3000-5000C)
- Basınç Kaynağı: ısı derecesi, metallerde **plastik** kıvam oluşturacak mertebede kalmışsa, bu tip kaynak basınç kaynağı olarak adlandırılır

Normal çelik yapılarda, genellikle ergitme kaynak kullanılır.

- Flux Welding: It is the welding that occurs by increasing the temperature of the metals to the melting point (enough to bring the metals to liquid consistency). (3000-5000C)
- Pressure Source: If the temperature remains at a level to form a plastic consistency in metals, this type of welding is called pressure welding.

In normal steel structures, flux welding is generally used.

Kaynak Türleri (types of weld)

- Elektrik arkı kaynakları
  - – Standart elektrik arkı kaynağı (elektrod kaynağı)
  - – Özlü elektrodlu elektrik arkı kaynağı
  - – Toz altı elektrik arkı kaynağı
  - – Gaz altı elektrik arkı kaynağı
    - WIG (wolfram inert gas) kaynağı
    - MIG (metal inert gas) kaynağı
    - MAG (metal active gas) kaynağı
- Gaz Kaynağı
- Lazer Işın Kaynağı
- Elektron Işın Kaynağı
- Sürtünme Kaynağı

- Electric arc Welding
  - Standard electric arc welding (electrode welding)- 3000˚C -5000 ˚C
  - Cored electrode electric arc welding
  - Submerged arc welding
  - Gas electric arc welding
    - WIG (wolfram inert gas) welding
    - MIG (metal inert gas) welding
    - MAG (metal active gas) welding
- Laser Beam Welding
- Electron Beam Welding
- Friction Welding

Kaynak makinesi

Kaynak Makineleri:

- *Kaynak jeneratörleri*
- *Kaynak redresörleri*
- *Kaynak transformatörleri*

Kaynak için elverişli elektrik akımının karakteristikleri 10~60 V ve 60~600A'dir.

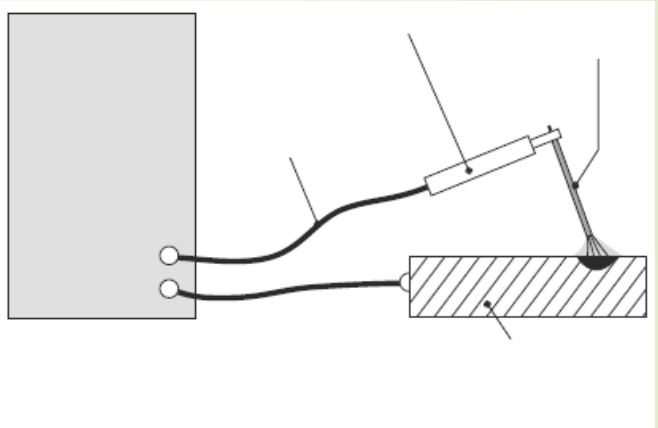
Elektrodlar: Sıvalı ve çıplak olmak üzere iki çeşit elektrod mevcuttur. Çıplak elektrodlar sadece önemsiz tespit dikişlerinde doğru akım ile uygulanabilir. Kuvvet aktaran kaynak dikişlerinde kullanılmamalıdır.

Welding Machines:

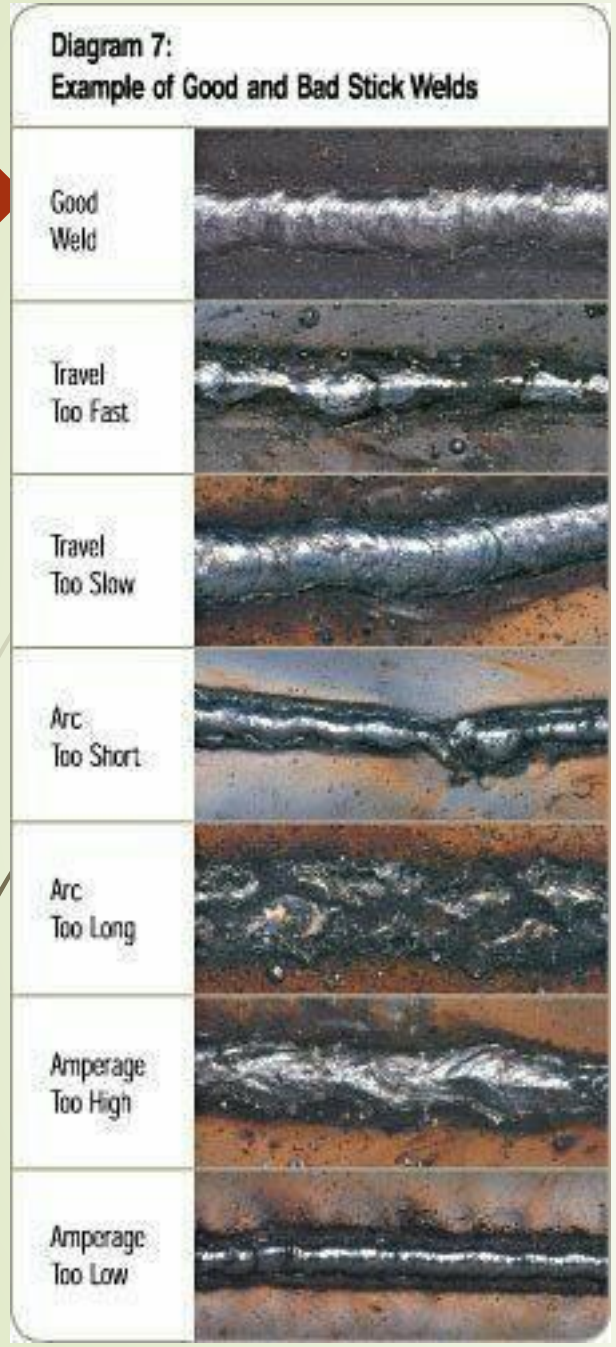
- Welding generators
- Welding rectifiers
- Welding transformers

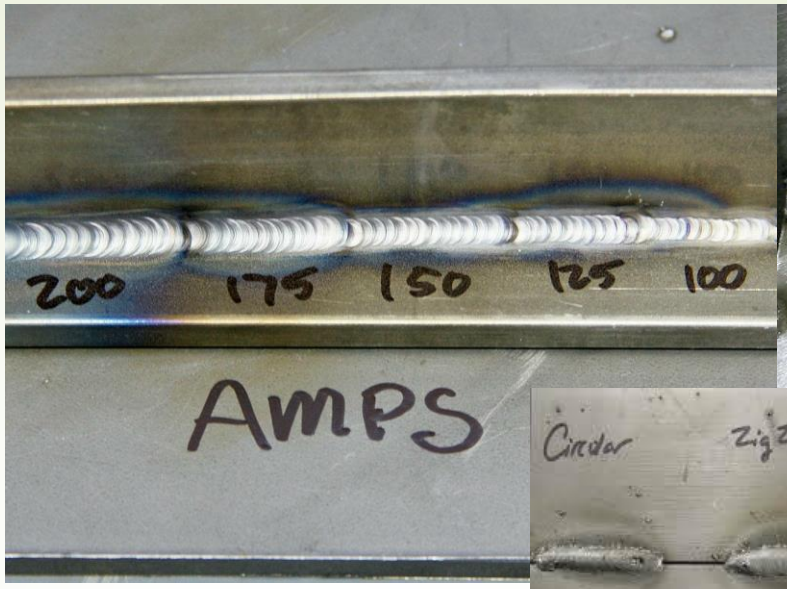
The characteristics of electric current suitable for welding are 10 ~ 60 V and 60 ~ 600A.

Electrodes: There are two types of electrodes, plastered and naked. Bare electrodes can only be applied in minor anchoring seams with direct current. It should not be used on force-transferring weld seams.

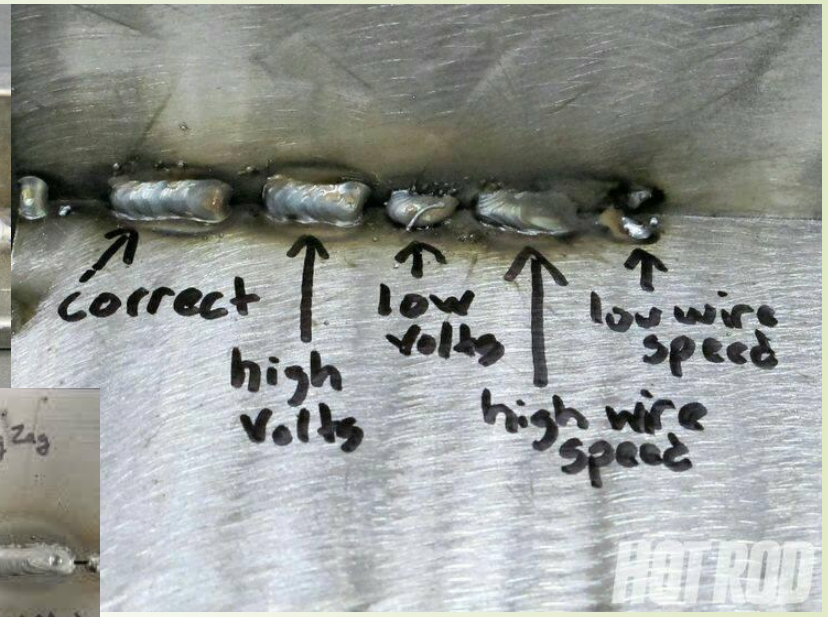


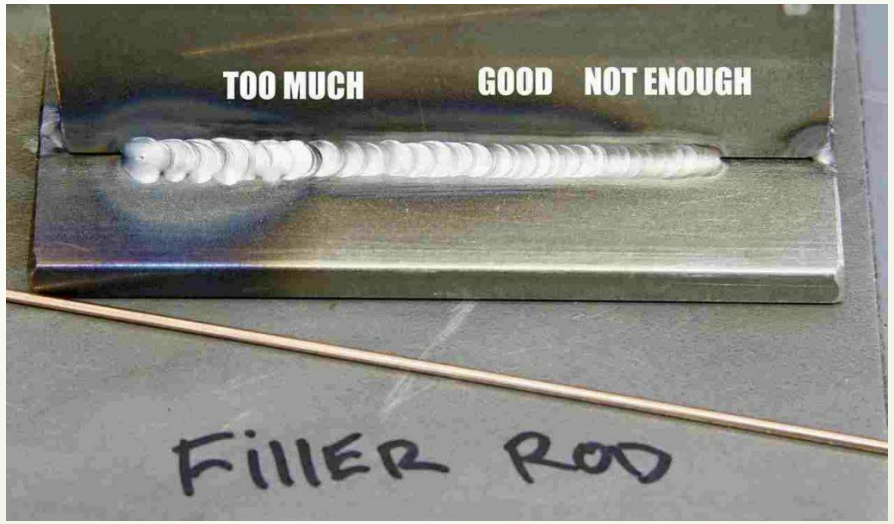


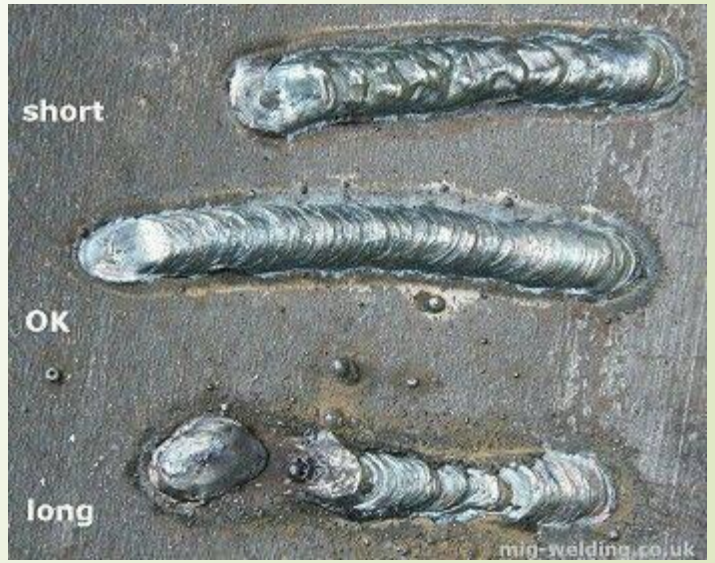

Diagram 7:
Example of Good and Bad Stick Welds
Good Weld
Travel Too Fast
Travel Too Slow
Arc Too Short
Arc Too Long
Amperage Too High
Amperage Too Low


200
175
150
125
100
AMPS


Circular
Zig Zag
MIG


correct
high volts
low volts
high wire speed
low wire speed
HOT ROD


TOO MUCH
GOOD
NOT ENOUGH
FILLER ROD


short
OK
long
mig-welding.co.uk


THERE ARE TWO
TYPES OF PEOPLE

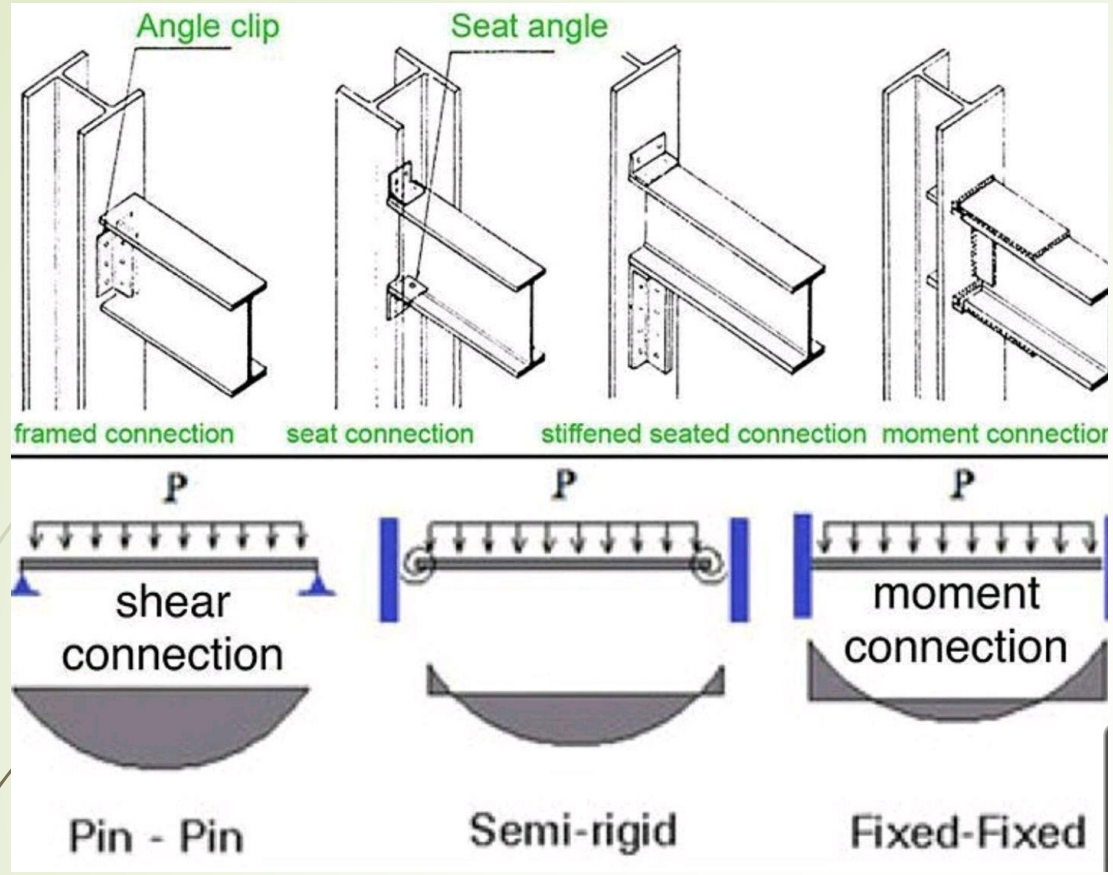

Amatour
Professional

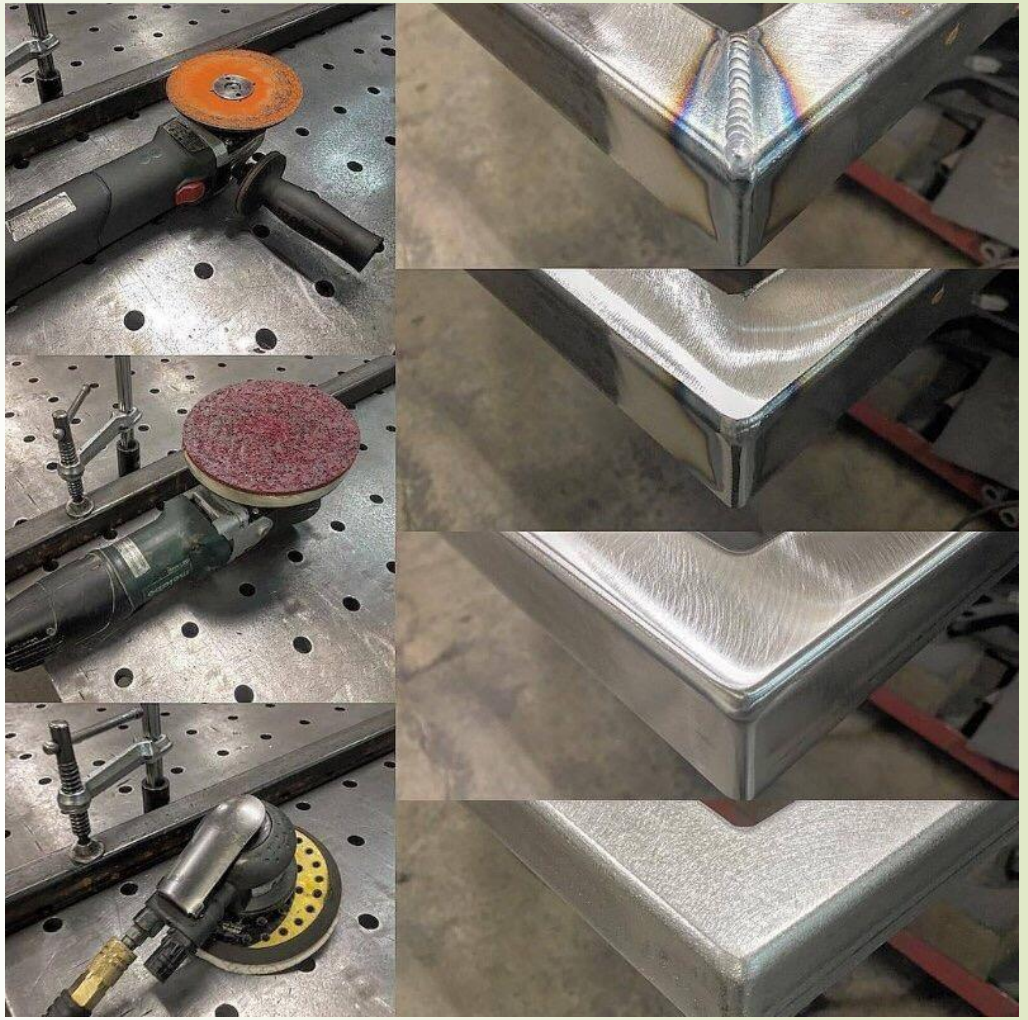


Kaynaklı birleşimlerin avantajları

- Kaynaklı birleşimler, bulonlu veya perçinli birleşimlerden **daha ekonomiktir**.
- Kaynağın uygulama alanı ,perçinli/bulonlu birleşimlerden daha fazladır.
- Kaynaklı birleşimlerde birleştirilen elemanlar direk olarak birbirlerine kaynakladığı için kaynaklı yapılar diğer birleşimlere göre daha rijittir.
- Kaynaklı yapılar gerçek anlamda sürekli/ devamlı (continuous) yapılardır ve daha temiz / düzenli görünürler
- Üretim ve montaj sırasında karşılaşılabilecek ufak hatalar kolaylıkla tolere edilebilir.
- Kaynaklı birleşimlerde daha az parça veya levha kullanıldığı için üretim daha kısa sürer

- Welded joints are more economical than bolted or riveted joints.
- The application area of welding is more than riveted / bolted joints.
- Welded structures are more rigid than other joints since the elements joined in welded joints are directly welded to each other.
- Welded structures are literally continuous structures and appear cleaner / tidy.
- Small errors that may be encountered during production and assembly can be easily tolerated.
- Production takes less time as fewer parts or plates are used in welded joints


Angle clip
Seat angle
framed connection
seat connection
stiffened seated connection
moment connection
P
P
P
shear connection
moment connection
Pin - Pin
Semi-rigid
Fixed-Fixed



Kaynaklı birleşimlerin dezavantajları

- Çok küçük moment aktarımı istenen basit birleşimlerde kaynaklı rijit birleşimler sorun yaratabilir.
- Kaynaklı birleşimlerde kalite kontrol işlemleri bulonlu/perçinli birleşimlere göre daha pahalıdır.
- Kaynaklı birleşim yapabilmek için özel ekipmana ihtiyaç vardır.
- Özellikle gaz kaynağında oluşabilecek kazalar ölümcül olabilir.
- Kaynaklı birleşimler hasar vermeden sökülemeyen birleşimlerdir.

- Welded rigid joints can cause problems in simple joints that require very small moment transfer.
- Quality control processes in welded joints are more expensive than bolted / riveted joints.
- Special equipment is required for welded joints.
- Accidents that may occur especially in the gas supply can be fatal.
- Welded joints are those that cannot be removed without damaging them.

Kaynak çeşitleri

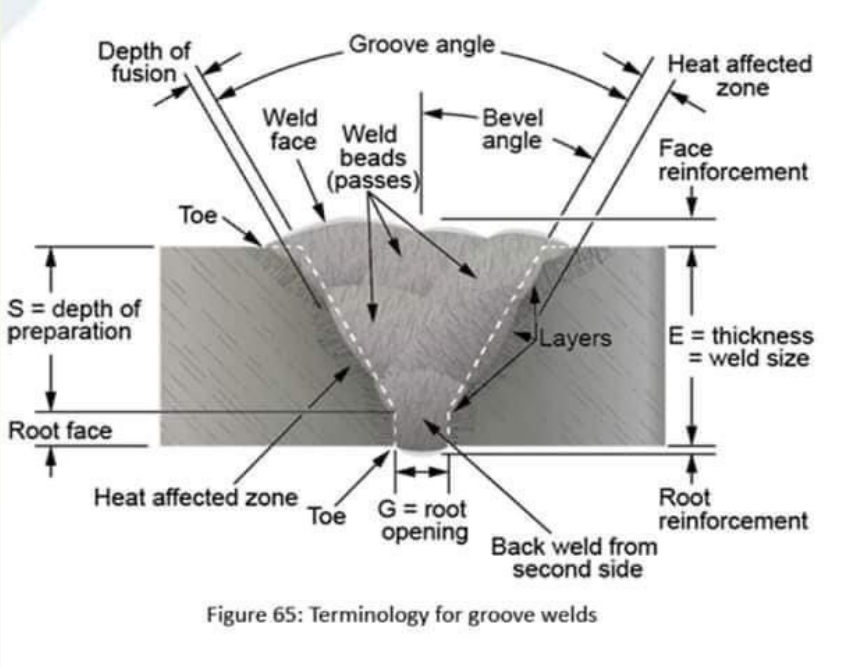
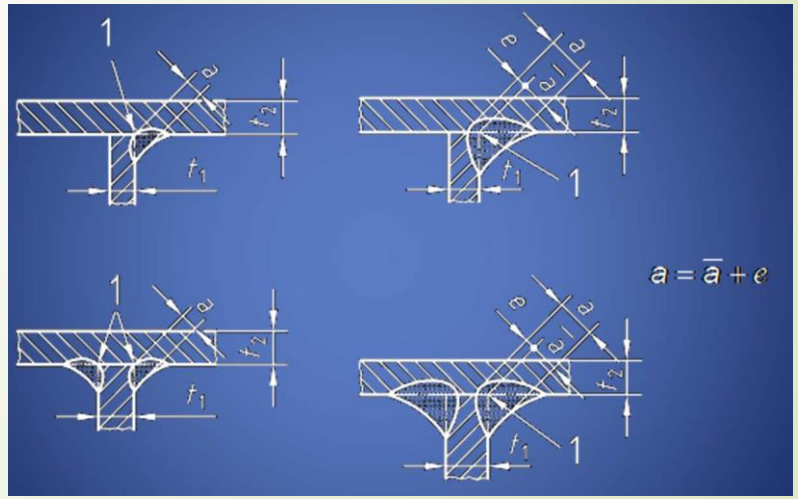
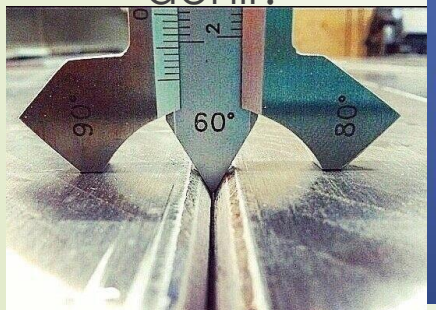


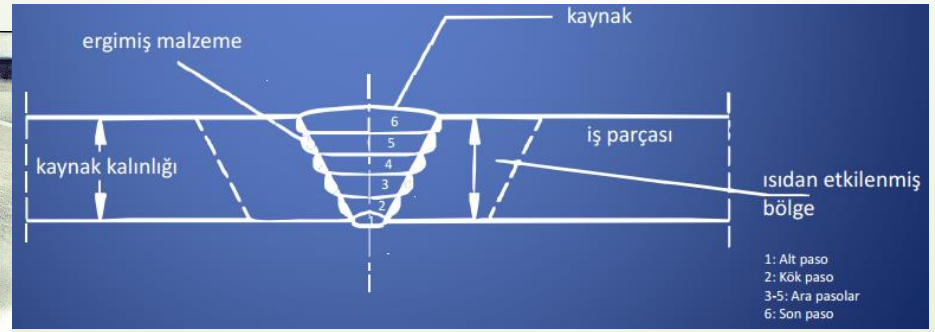
Figure 65: Terminology for groove welds

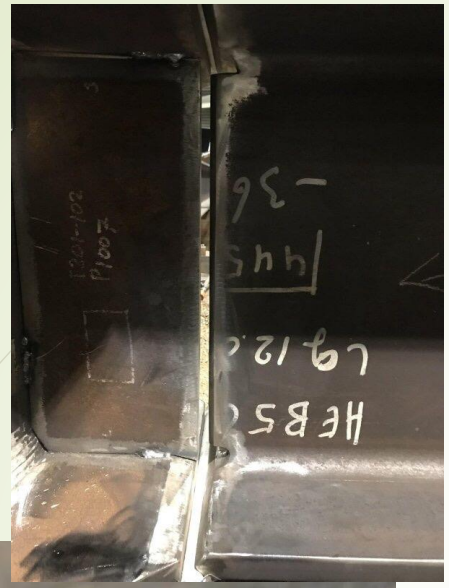
Ergitme kaynağı yöntemleriyle çekilen kaynak dikişleri başlıca iki çeşittir:

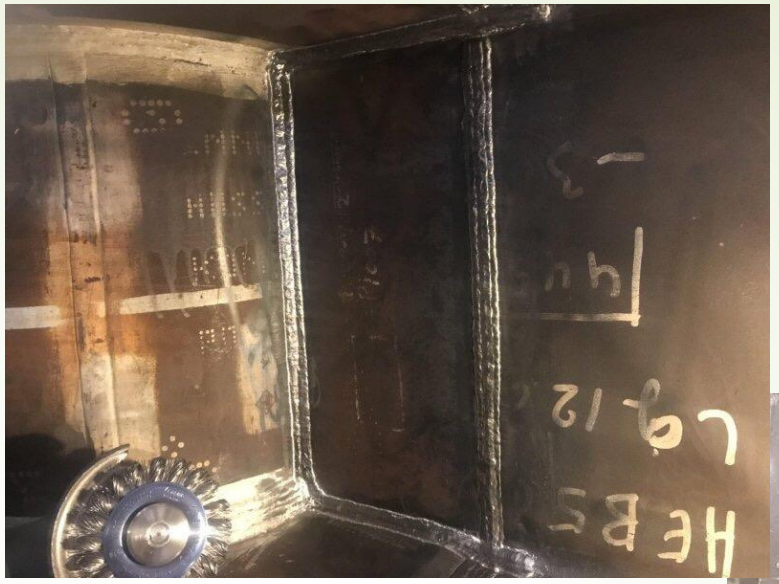
- **Küt kaynak dikişleri**: Ayni düzlemde bulunan iki levhanın yan yana getirilen kenarları boyunca çekilen kaynak dikişleri küt kaynak dikişleridir. Levha kenarlarının işleniş biçimlerine göre özel isimler alırlar.
- **Köşe kaynak dikişleri**: İki çelik elemanın birbirine dik veya en az 60 teşkil eden yüzeyleri arasındaki köşelere çekilen dikişlere köşe kaynak dikişleri denir.













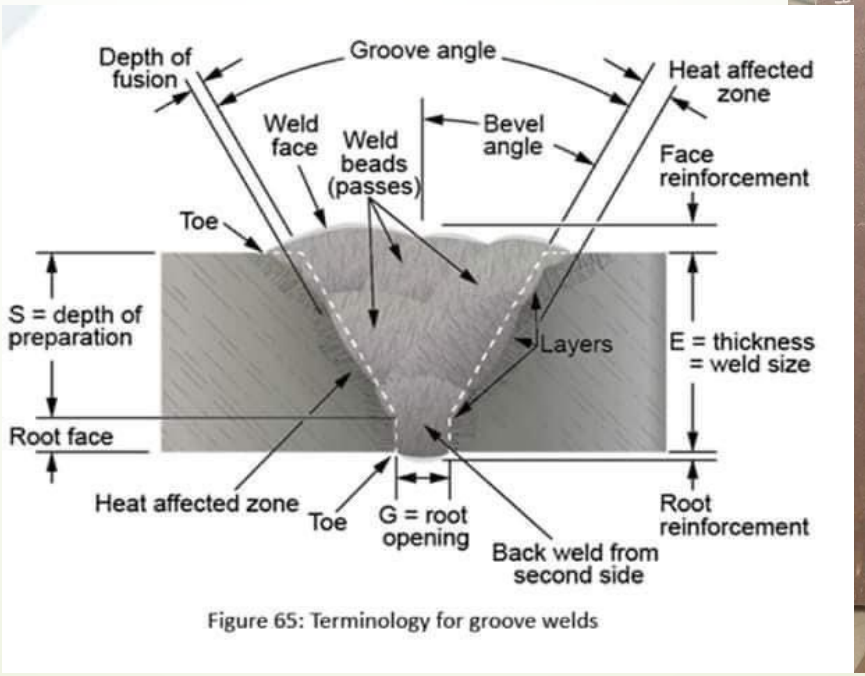
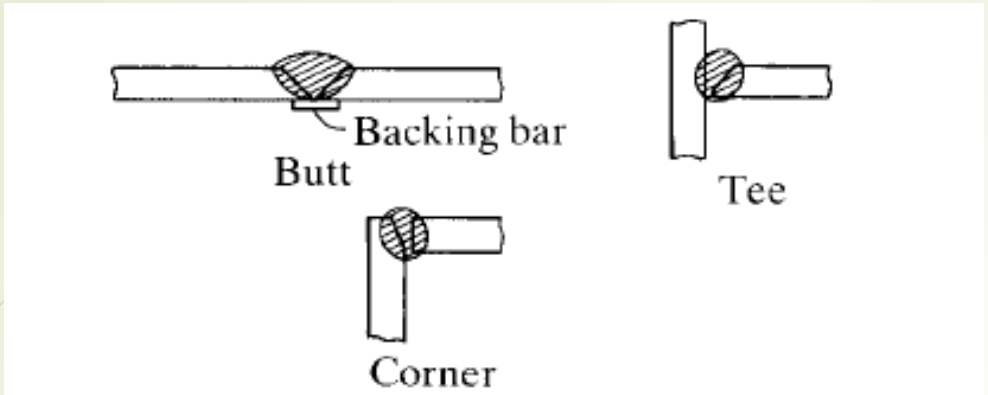
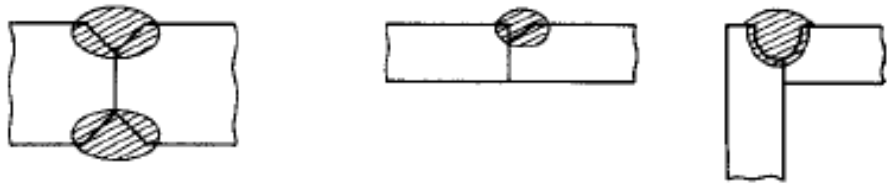


Figure 65: Terminology for groove welds

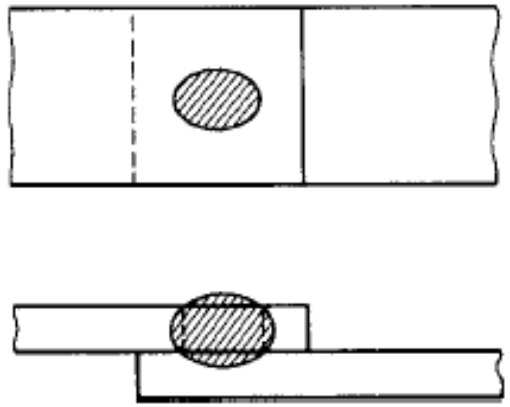




(a) Complete penetration groove welds



(b) Partial penetration groove welds



Dikiş türleri

Dikiş türleri	Gösterim	Sembol
V – Dikişi		V
Yarım V – Dikişi		V
Çift V – Dikişi		X
Çift Yarım V – Dikişi (K)		K
Y – Dikişi		Y
Yarım Y – Dikişi		Y
Çift Y – Dikişi		X
Çift Yarım Y – Dikişi		K
Köşe Kaynak Dikişi		◺
Çift Köşe Kaynak Dikişi		◺

Tam Penetrasyonlu

Kısmı Penetrasyonlu

Küt Kaynak

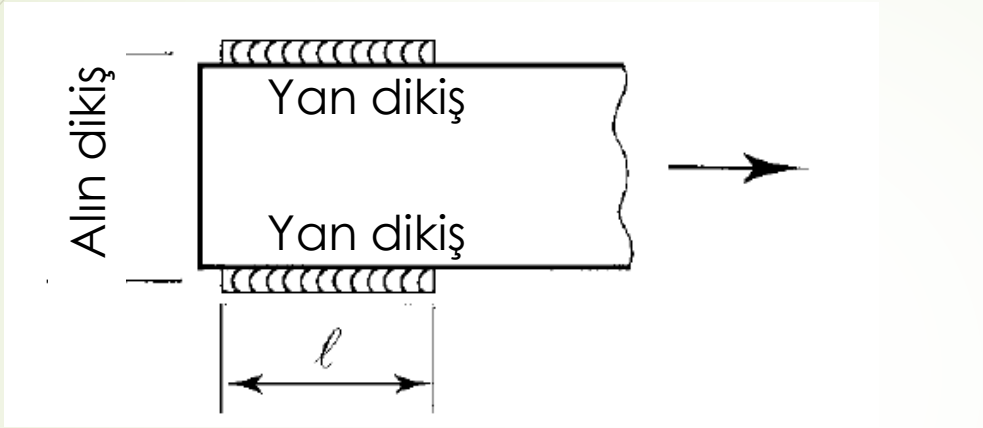
CJP

PJP

Köşe Kaynak

Fillet Welds

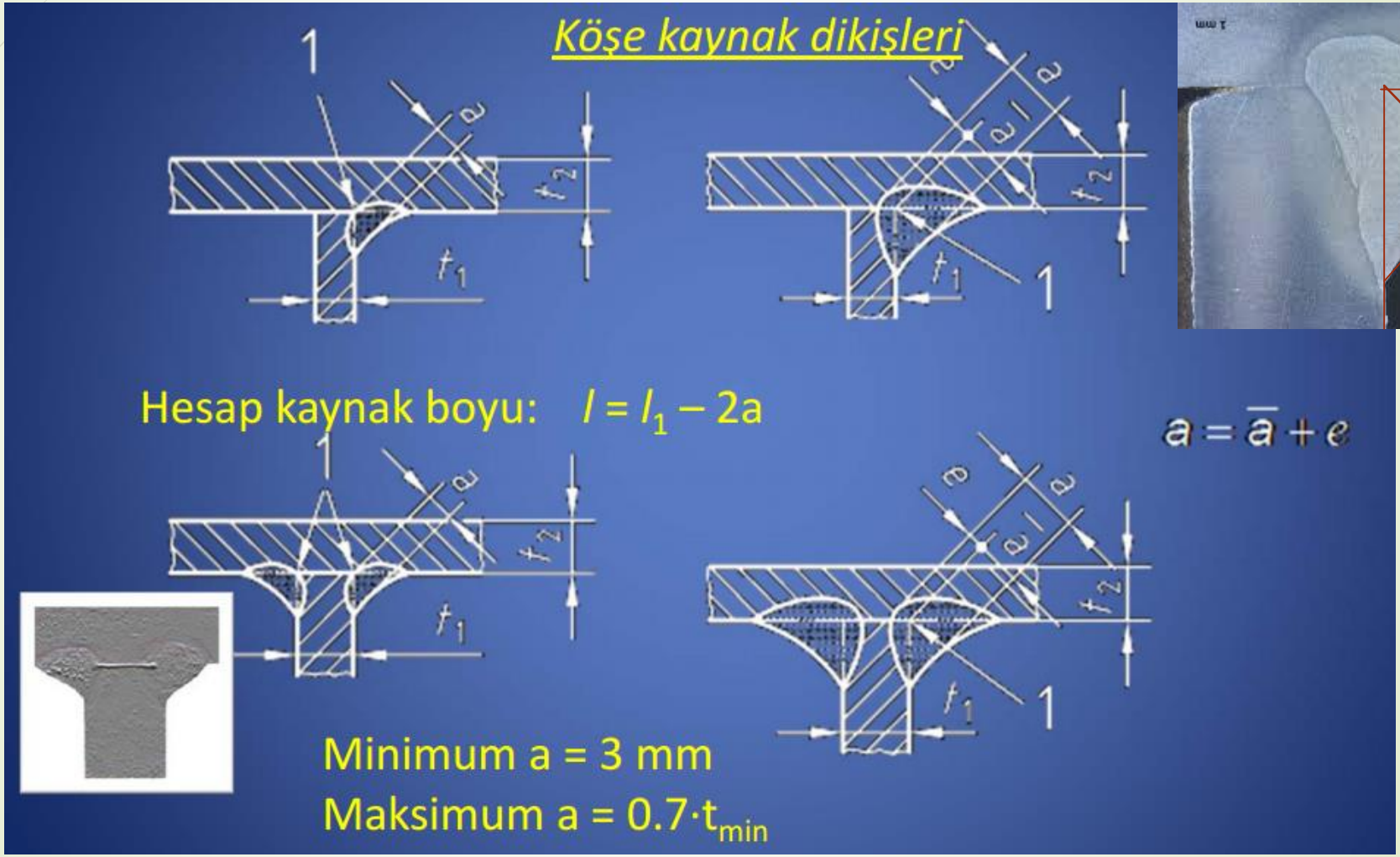
Köşe Kaynak (Fillet Welds)

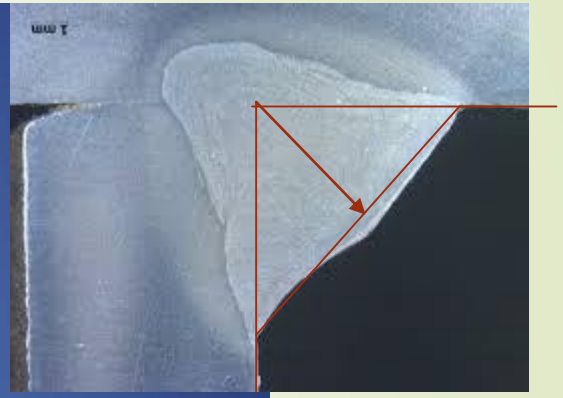




- Köşe kaynağı dikişi doğrultusunda kuvvete paralel ise yan dikiş dik ise alın dikişi adını alır.
- Elemanların birbiri üzerine bindirilmesine izin verilen durumlarda köşe kaynağı kullanılır.
- İki elemanın birbirine dik açıları dik veya en az 60` oluşturan yüzeylere çekilen kaynak dikişidir.

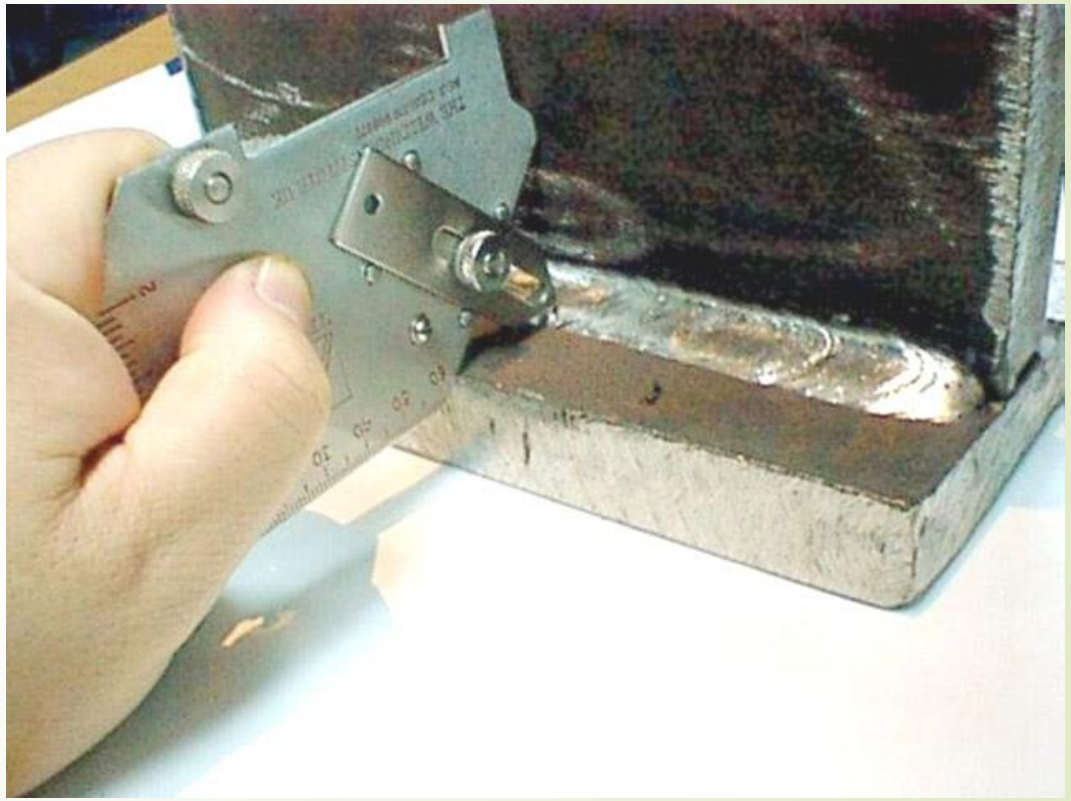
Kaynak bilgileri





Kaynak kalınlığı ölçmek

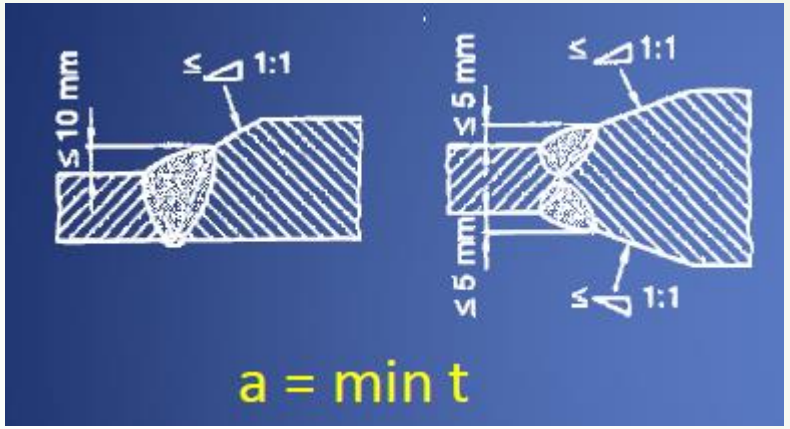


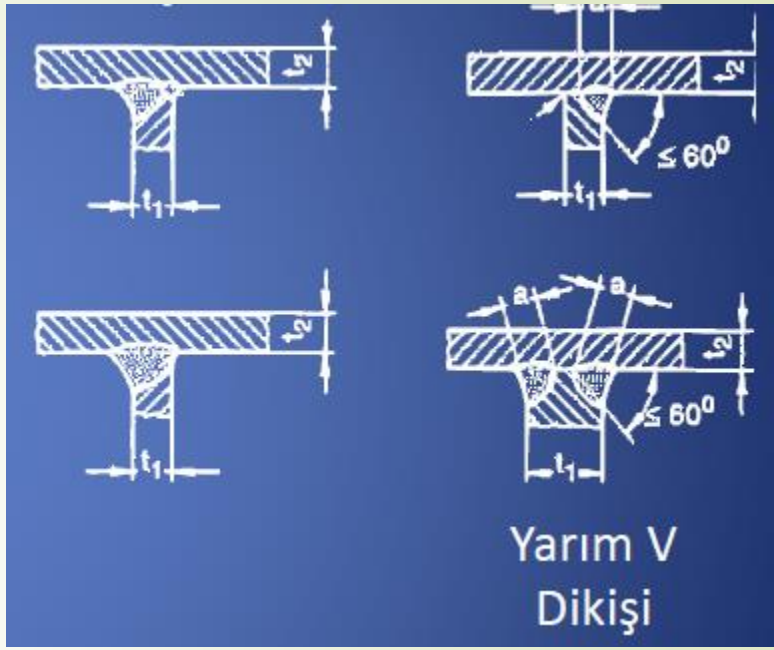


Kaynak kalınlığı

Küt kaynak da

- $a=t_{min}$



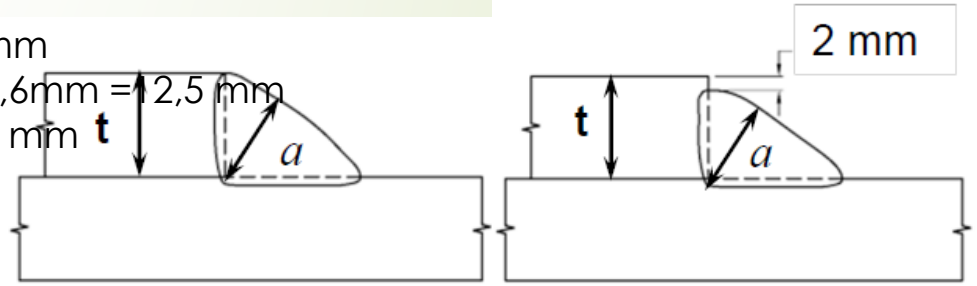


Birleşen İnce Elemanın Kalınlığı, t [mm]	Minimum Köşe Kaynak Kalınlığı,[a] [mm]
$6 \geq t$	3.0
$13 \geq t > 6$	3.5
$19 \geq t > 13$	4.0
$38 \geq t > 19$	5.5

[a]: Köşe kaynakların kalınlığı. Tek geçişli kaynaklar kullanılmalıdır.
Not: Köşe kaynakların maksimum kalınlığı için **Bölüm 13.2.2.2** ye bakınız.

t=15mm amx=9mm
t=20mm amx=12,6mm =12,5 mm
t=7 mm amx=3,5 mm
t=6 mm amx=

- t<6mm ise a_{max}=0,7*t (mm)
- t>6mm ise a_{max}=0,7*(t -2) mm



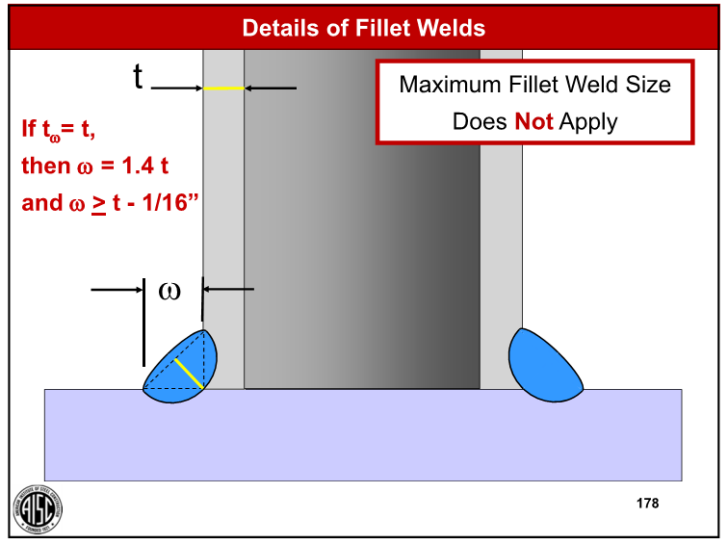
$t \leq 6$ mm
$a_{maks} = 0.7t$

$t > 6$ mm
$a_{maks} = 0.7(t - 2\text{ mm})$

Köşe kaynakda maksimum kalınlık

3.5.2 Maximum Size of Fillet Welds

Both the AISC Specification and AWS D1.1 have provisions that specify maximum fillet weld sizes. These provisions are frequently misunderstood and misapplied. AISC Specification Section J2.2b requires a maximum fillet weld size along the edges of material ¼ in. or more in thickness. Note that this applies only to the situation where welds are applied "along the edges." Such conditions include lap joints, and some corner joint configurations, but do not include T-joints (Figure 3–12).



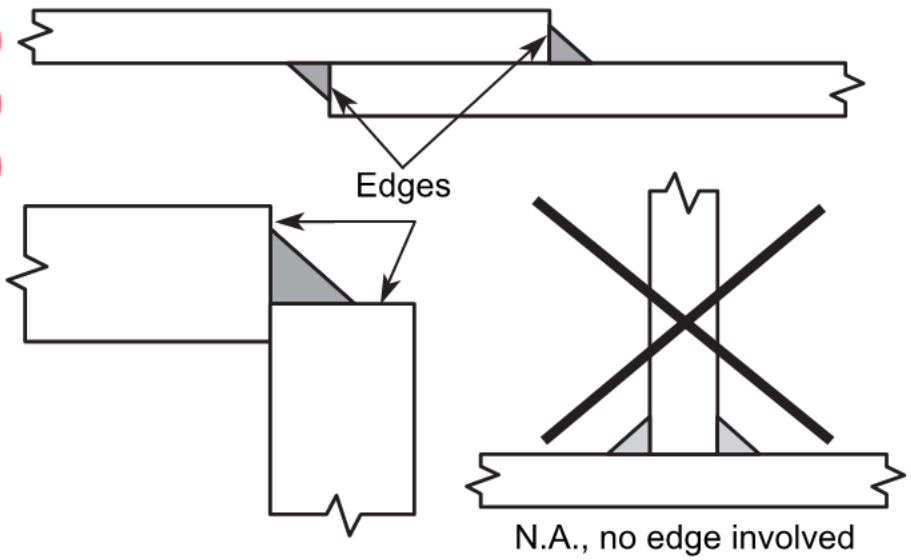
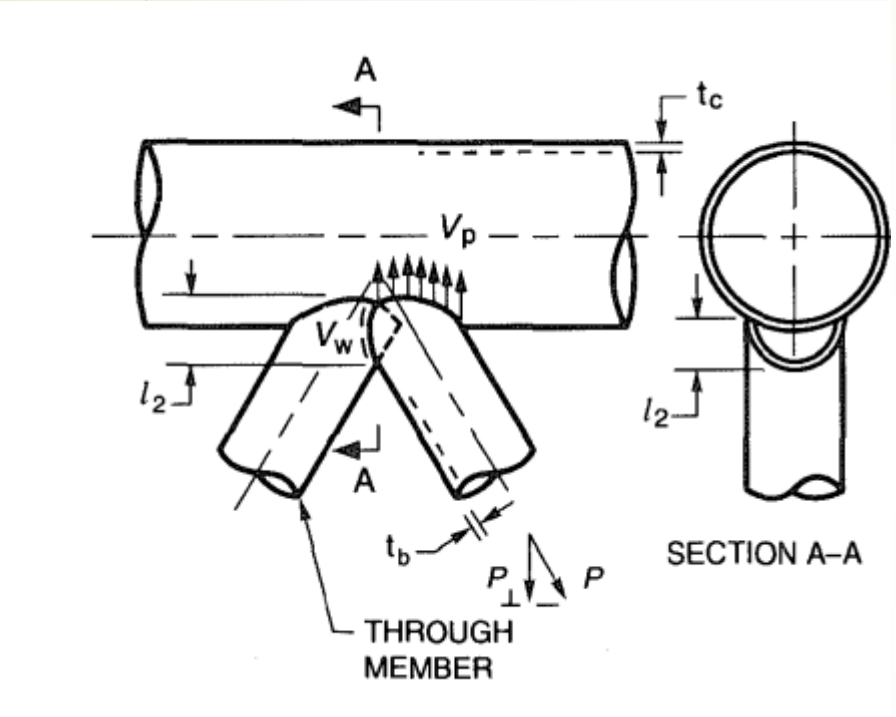
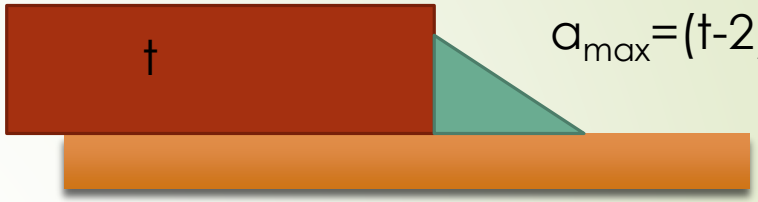


Figure 3–12. Welds on edges.

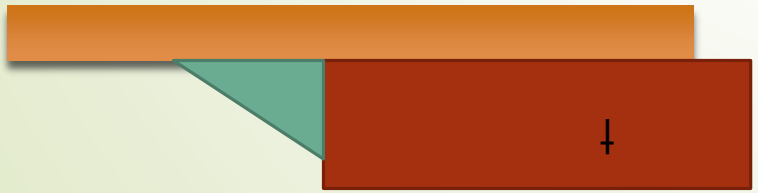
- 2.40.1.6 Overlapping Connections.
  (6) Minimum weld size for fillet welds shall provide effective throat of 1.0 th for Fy < 40 ksi (280 MPa), 1.2 th for Fy > 40 ksi (280 MPa).

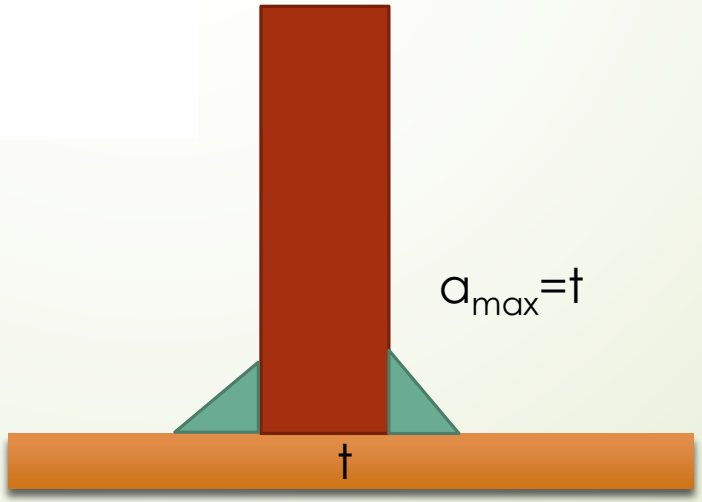


Maksimum thickness of weld
t>6 mm
a_{max}=(t-2)*0,7

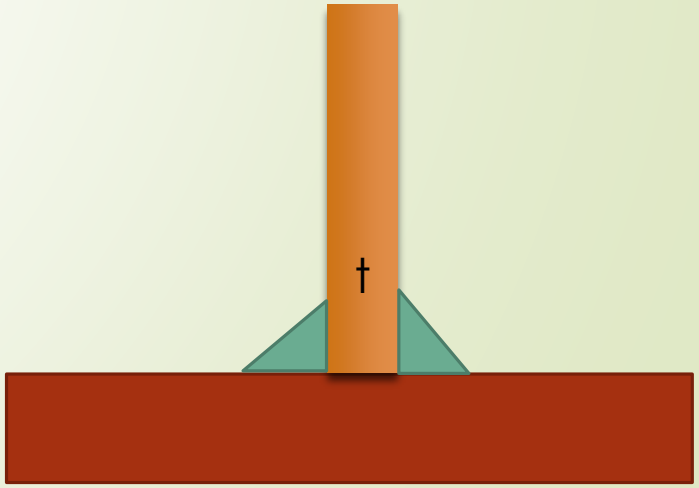


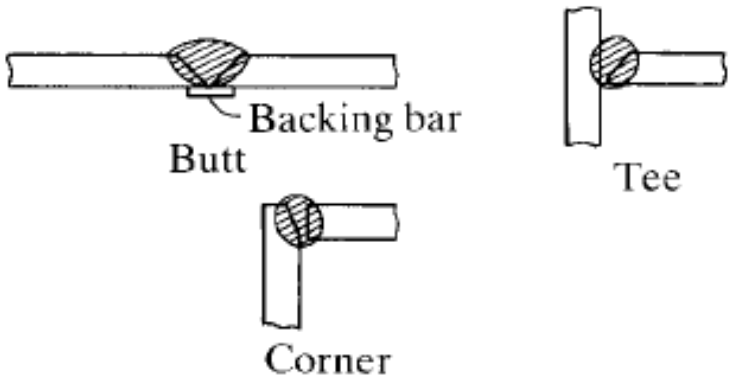




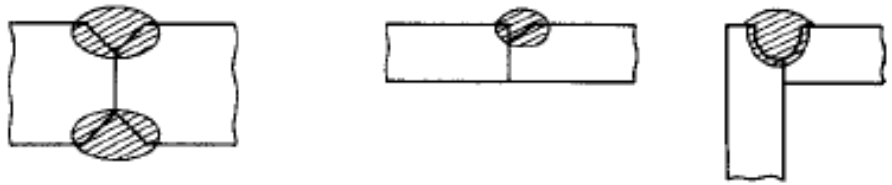


a_{max}=t

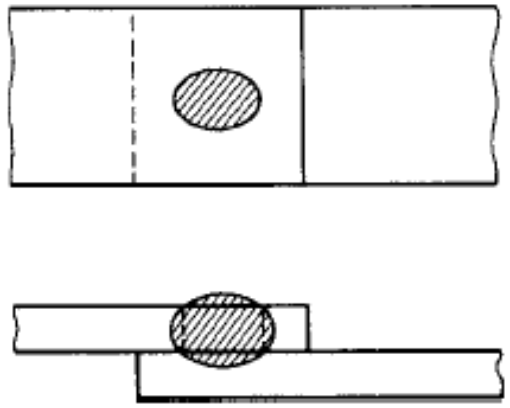




(a) Complete penetration groove welds



(b) Partial penetration groove welds



Minimum kısmı penetrasyonlu küt kaynak kalınlığı

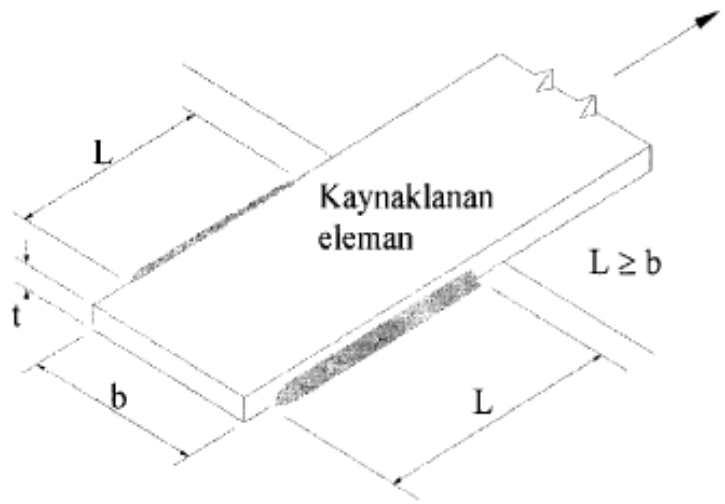
Birleşen İnce Elemanın Kalınlığı, t [mm]	Minimum Etkin Kalınlık,[a] [mm]
$6 \geq t$	3
$13 \geq t > 6$	5
$19 \geq t > 13$	6
$38 \geq t > 19$	8
$57 \geq t > 38$	10
$150 \geq t > 57$	13
$t > 150$	16

[a]: **Tablo 13.1** e bakınız.

welding shall be not less than four times the weld size, with a minimum of $1\frac{1}{2}$ in. (38 mm).

Dikiş uzunluğu

Köşe kaynakların minimum etkin uzunluğu, kaynak kalınlığının 6 katından veya 40mm den az olamaz. Bu koşulun sağlanamadığı durumda, kaynağın etkin kalınlığı, kaynak uzunluğunun 1/6 sı olarak gözönüne alınacaktır. Lamadan teşkil edilen bir çelik çekme elemanın uç birleşiminde sadece boyuna doğrultuda köşe kaynaklar kullanılması halinde, bu kaynakların her birinin uzunluğu kaynaklar arası dik uzaklıktan daha küçük olamaz, (**Şekil 13.3**).



Şekil 13.3 – Sadece boyuna kaynak uygulanan uç birleşimi

Elemanların kaynaklı uç birleşimlerinde etkin kaynak uzunluğu aşağıdaki koşullar dikkate alınarak hesaplanacaktır.

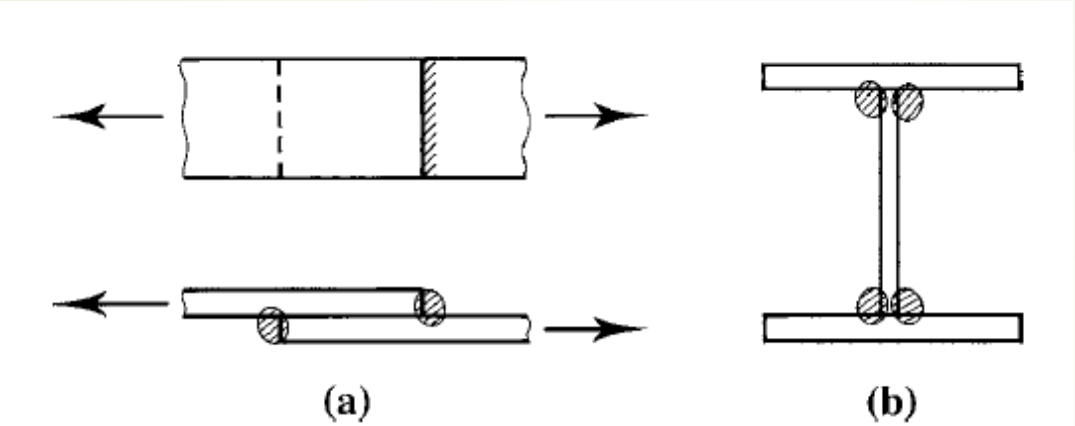
$L \le 150a$ için $L_e = L$

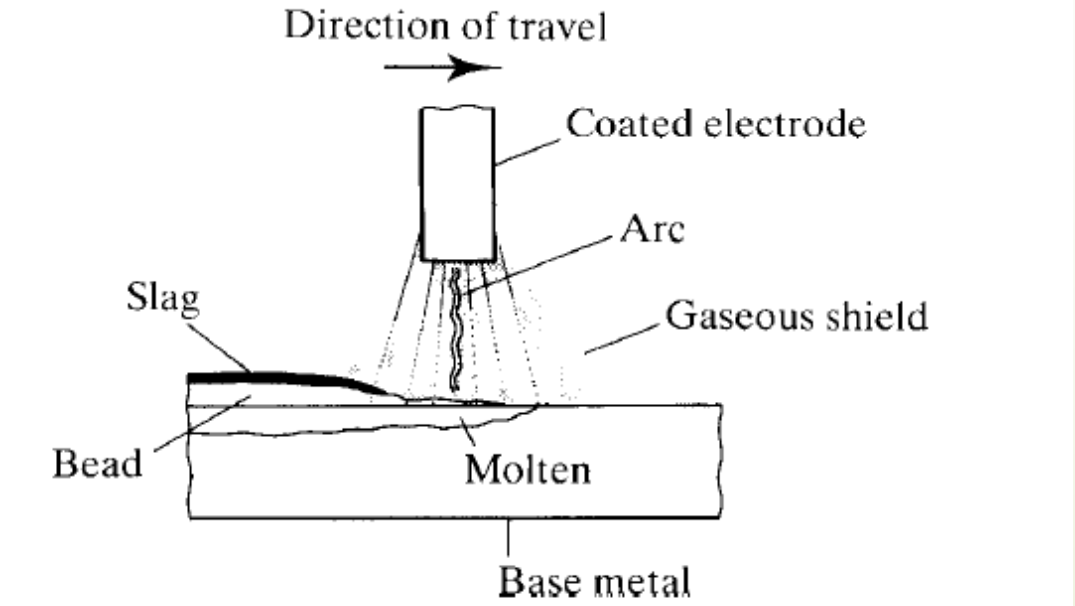
$150a < L \le 400a$ için $L_e = \beta L$ $\qquad \beta = 1.2 - 0.0014(L/a) \le 1.0$

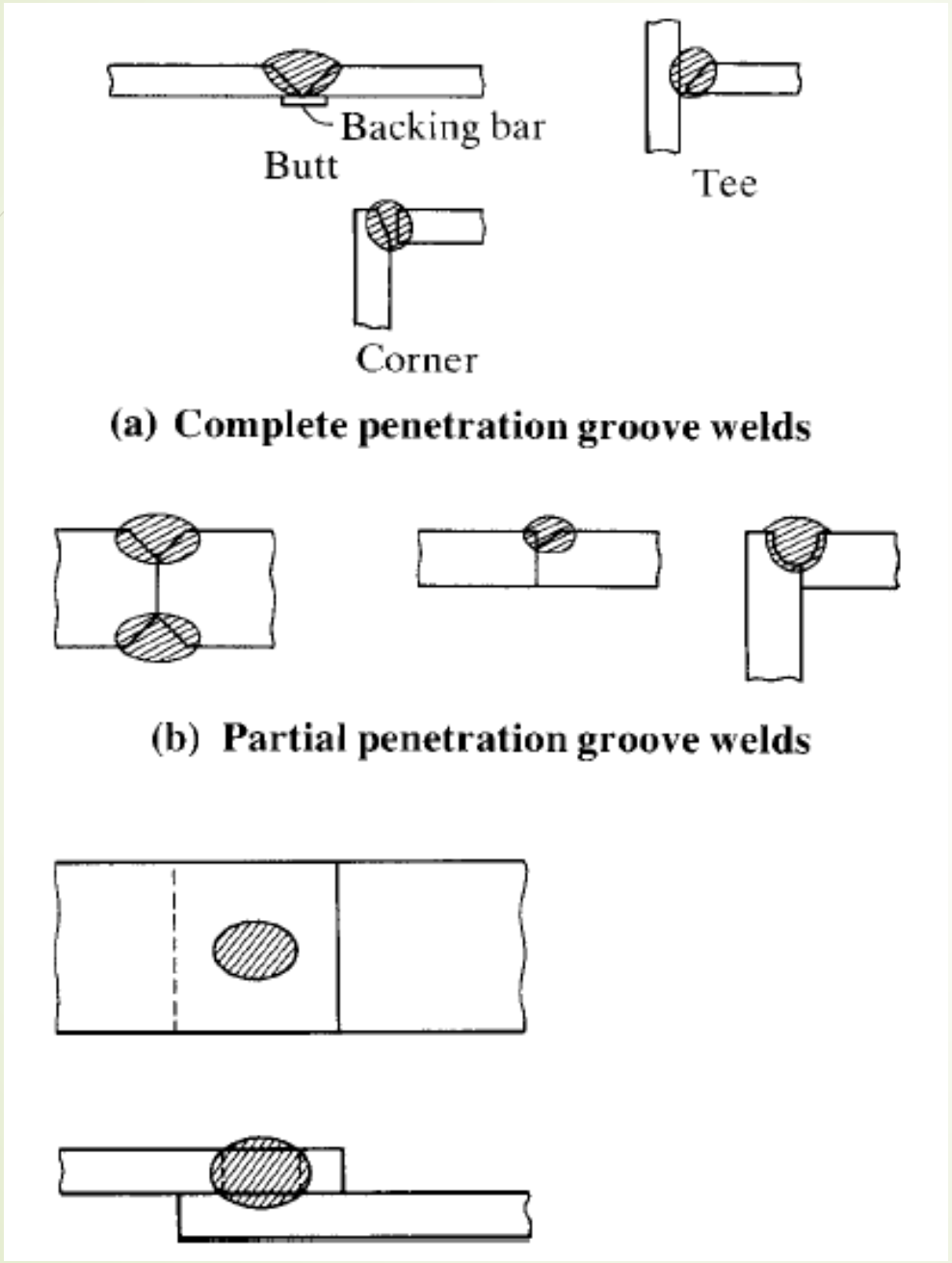
$400a < L$ için $L_e = 250a$

Min Length Weld
a=6mm 6*6=36
6*150=900 mm
40 mm
40 mm min length
a=8 mm 8*6=48mm
8*150=1200 mm
40 mm
50 mm min length
a=3mm 3*6=18 mm
40 mm
40 min length

Kaynak


(a)
(b)


Direction of travel
Coated electrode
Arc
Slag
Gaseous shield
Bead
Molten
Base metal


Backing bar
Butt
Tee
Corner
(a) Complete penetration groove welds
(b) Partial penetration groove welds

Köşe Kaynak

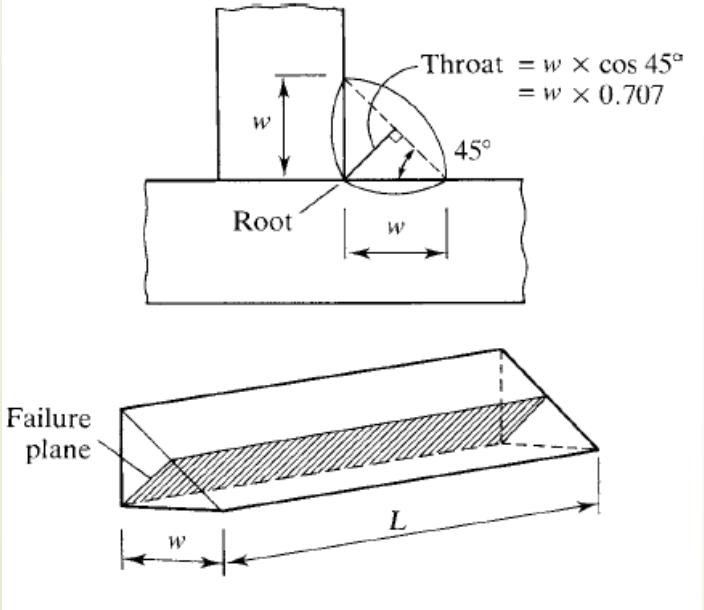
$$R_{nBM} = F_{nBM} A_{BM}$$

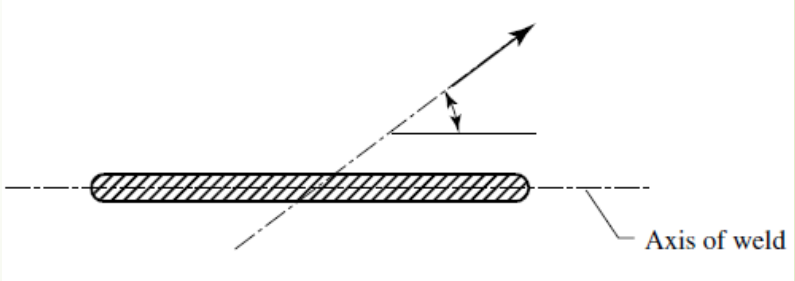
$$R_{nw} = F_{nw} A_{we}$$



$$F_{nw} = 0.60 F_E$$

$$F_{nw} = 0.60 F_E \left(1.0 + 0.50 \sin^{1.5} \theta\right)$$





Direction of load (θ)	$F_{nw} = 0.60F_{EXX}(1.0 + 0.50 \sin^{1.5}\theta)$
0°	$0.60F_{EXX}$ (1.0)
15°	$0.60F_{EXX}$ (1.066)
30°	$0.60F_{EXX}$ (1.177)
45°	$0.60F_{EXX}$ (1.297)
60°	$0.60F_{EXX}$ (1.403)
75°	$0.60F_{EXX}$ (1.475)
90°	$0.60F_{EXX}$ (1.5)

Electrode Strength Coefficient,C1			
Electrode	Fexx (ksi)	Fexx (MPa)	C1
E60	60	410	0.857
E70	70	480	1.00
E80	80	550	1.03
E90	90	620	1.16
E100	100	690	1.21
E110	110	760	1.34

Base Metal		Matching Filler Metal
A36 $\leq$ 3/4 in. thick		60 & 70 ksi Electrodes
A36 > 3/4 in. A588* A1011	A572 (Gr. 50 & 55) A913 (Gr. 50) A992 A1018	SMAW: E7015, E7016, E7018, E7028 Other processes: 70 ksi electrodes
A913 (Gr. 60 & 65)		80 ksi electrodes

* For corrosion resistance and color similar to the base see AWS D1.1, Sect. 3.7.3

Notes:

1. Electrodes shall meet the requirements of AWS A5.1, A5.5, A5.17, A5.18, A5.20, A5.23, A5.28 and A5.29.
2. In joints with base metals of different strengths use either a filler metal that matches the higher strength base metal or a filler metal that matches the lower strength and produces a low hydrogen deposit.

Table 8–2 (continued)
Prequalified Welded Joints

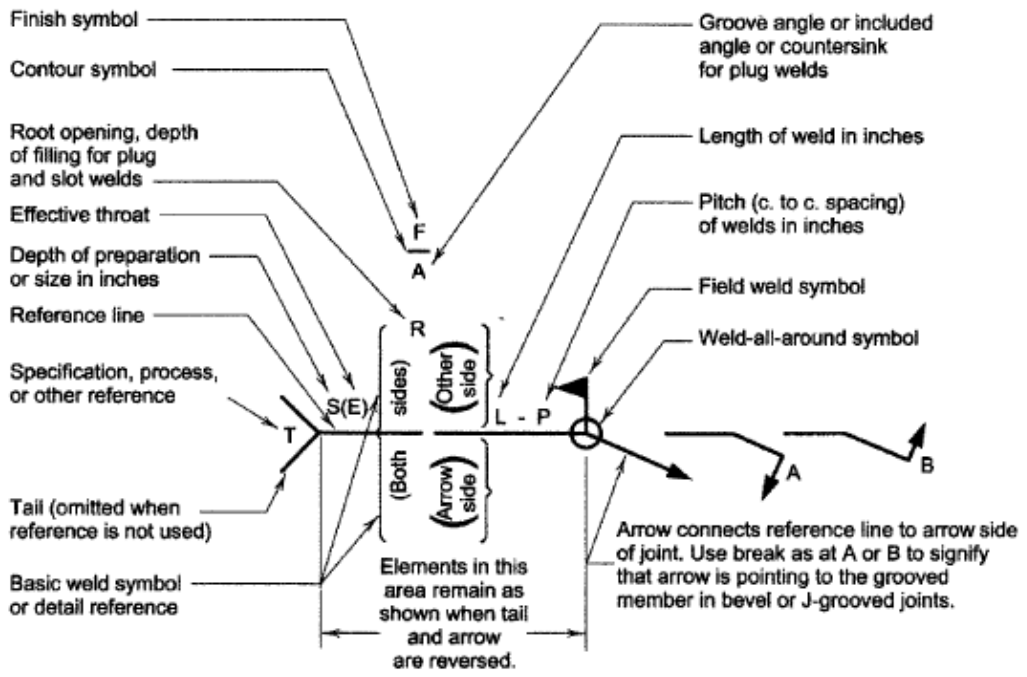
Basic Weld Symbols

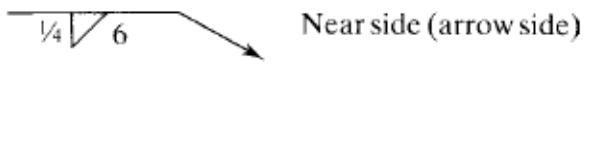
Back	Fillet	Plug or Slot	Groove or Butt						
			Square	V	Bevel	U	J	Flare V	Flare Bevel

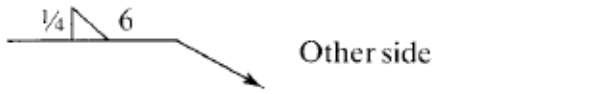
Supplementary Weld Symbols

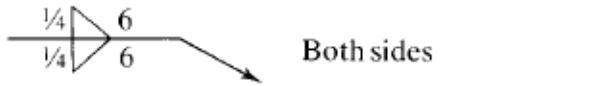
Backing	Spacer	Weld All Around	Field Weld	Contour		For other basic and supplementary weld symbols, see AWS A2.4
				Flush	Convex	

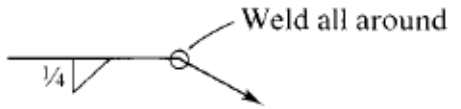
Standard Location of Elements of a Welding Symbol

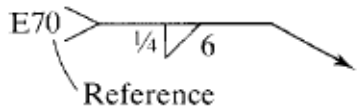


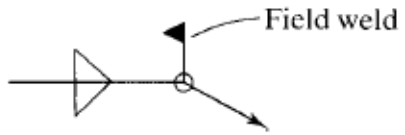












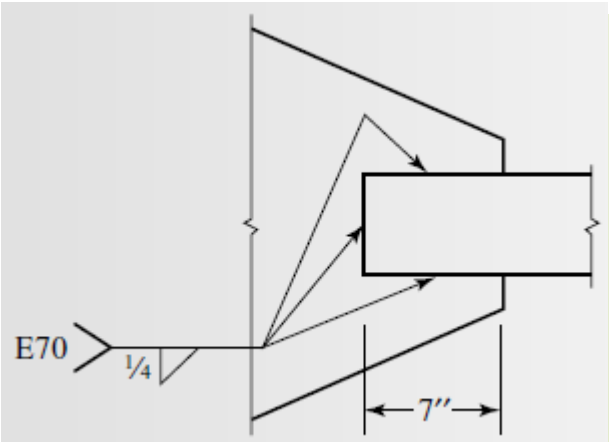
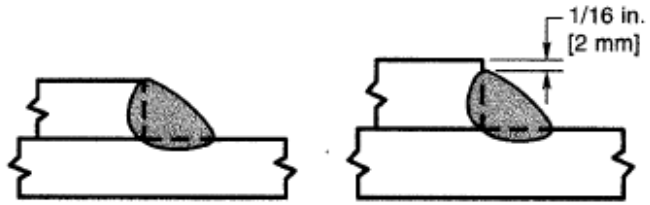


Table 8–2 (continued)

Prequalified Welded Joints
Fillet Welds

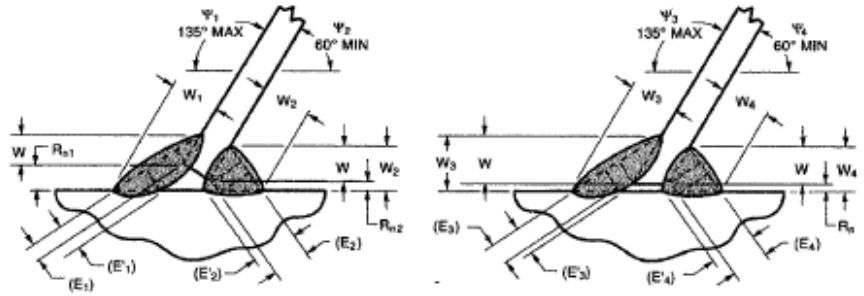
FILLET

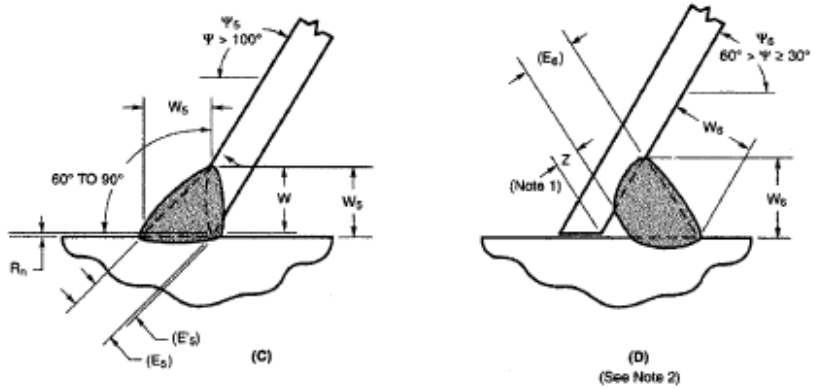


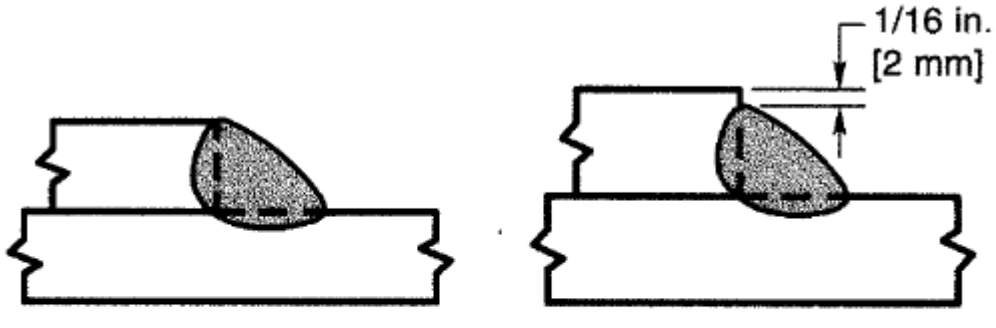
BASE METAL LESS THAN 1/4 in. [6 mm] THICK (A)

BASE METAL 1/4 in. [6 mm] OR MORE IN THICKNESS (B)

MAXIMUM DETAILED SIZE OF FILLET WELD ALONG EDGES



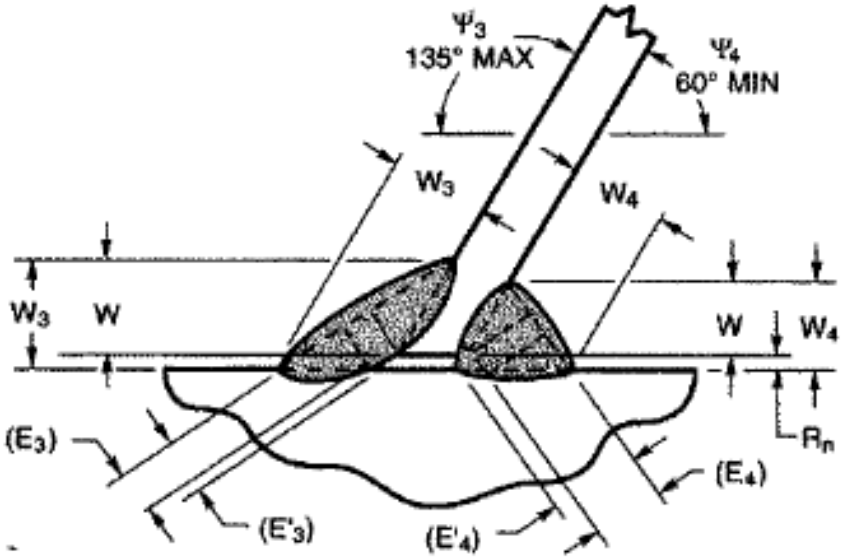


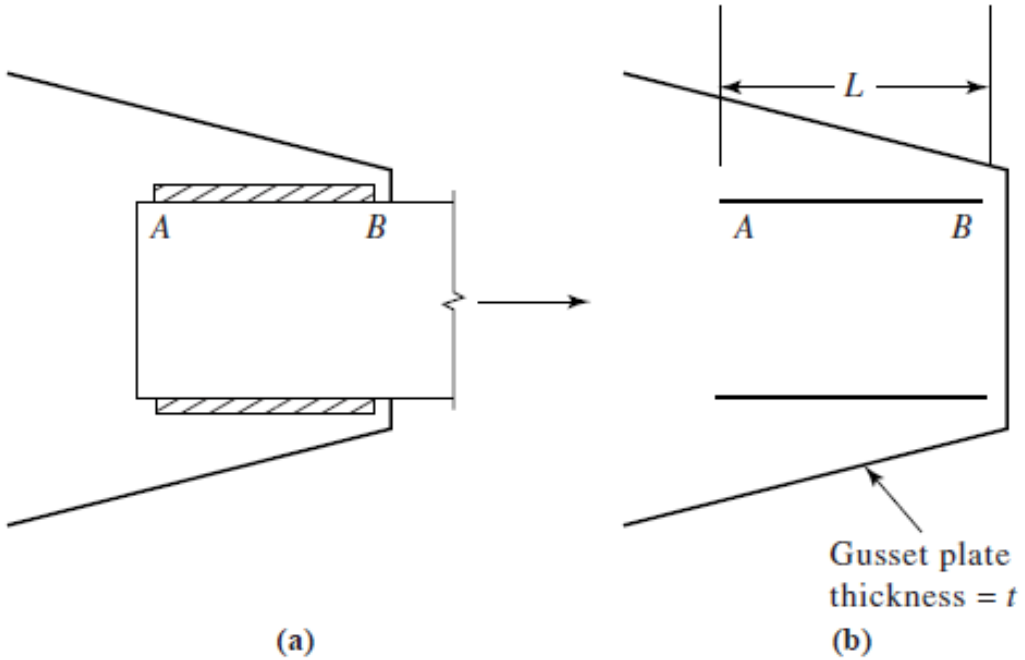


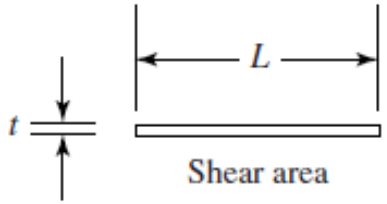
BASE METAL LESS THAN 1/4 in. [6 mm] THICK (A)

BASE METAL 1/4 in. [6 mm] OR MORE IN THICKNESS (B)

MAXIMUM DETAILED SIZE OF FILLET WELD ALONG EDGES







LRFD Equations

Weld shear strength:

$$\phi R_n = 0.75(0.707wF_{nw}) \text{ for a one-inch length} \qquad (7.29)$$

Base-metal shear strength:

$$\phi R_n = \min[1.0(0.6F_y t), 0.75(0.6F_u t)] \text{ for a one-inch length} \qquad (7.30)$$

TABLO 13.5 – KAYNAKLI BİRLEŞİMLERİN (YDKT) VEYA (GKT) ESASLARINA GÖRE TASARIM VE GÜVENLİ DAYANIMLARI

Yük Tipi ve Kaynak Eksenine Göre Doğrultusu	İlgili Metal	ϕ ve Ω	Karakteristik Gerilme (F_{nBM} veya F_{nw})	Etkin Alan (A_{BM} veya A_{we})	Kaynak Metali Gerekli Dayanım Düzeyi
TAM PENETRASYONLU KÜT KAYNAKLAR					
Kaynak eksenine dik çekme	Düğüm noktasının dayanımında esas metal belirleyicidir.				**Bölüm 13.2.6**
Kaynak eksenine dik basınç	Düğüm noktasının dayanımında esas metal belirleyicidir.				**Bölüm 13.2.6**
Kaynak eksenine paralel çekme ve basınç	Kaynak eksenine paralel çekme ve basınç etkilerinin düğüm noktası tasarımında göz önüne alınmasına gerek yoktur.				**Bölüm 13.2.6**
Kesme	Düğüm noktasının dayanımında esas metal belirleyicidir.				**Bölüm 13.2.6**

KISMİ PENETRASYONLU KÜT KAYNAKLAR, ENKESİTİNİN BİR KENARI EĞRİSEL VE ENKESİTİNİN İKİ KENARI EĞRİSEL KÜT KAYNAKLAR					
Kaynak eksenine dik çekme	Esas Metal	$\phi = 0.75$ $\Omega = 2.00$	F_u	**Bölüm 13.4**	**Bölüm 13.2.6**
	Kaynak Metali	$\phi = 0.80$ $\Omega = 1.88$	$0.60F_E$	**Bölüm 13.2.1.1**	
Bölüm13.1.4(a)'ya göre tasarlanan kolon-taban levhası bağlantıları ve kolon eklerinde basınç	Basınç etkisinin kaynak tasarımında göz önüne alınmasına gerek yoktur.				
Bölüm13.1.4(b)'de tanımlanan ve kolonlar hariç birleşen elemanlar arasında tam temasın sağlandığı birleşimlerde basınç	Esas Metal	$\phi = 0.90$ $\Omega = 1.67$	F_y	**Bölüm 13.4**	
	Kaynak Metali	$\phi = 0.80$ $\Omega = 1.88$	$0.60F_E$	**Bölüm 13.2.1.1**	
Birleşen elemanlar arasında tam temasın sağlanmadığı birleşimlerde basınç	Esas Metal	$\phi = 0.90$ $\Omega = 1.67$	F_y	**Bölüm 13.4**	
	Kaynak Metali	$\phi = 0.80$ $\Omega = 1.88$	$0.90F_E$	**Bölüm 13.2.1.1**	
Kaynak eksenine paralel çekme ve basınç	Kaynak eksenine paralel çekme ve basınç etkilerinin kaynak tasarımında göz önüne alınmasına gerek yoktur.				
Kesme	Esas Metal	**Bölüm 13.4**			
	Kaynak Metali	$\phi = 0.75$ $\Omega = 2.00$	$0.60F_E$	**Bölüm 13.2.1.1**	

KÖŞE KAYNAKLAR					
Yük Tipi ve Kaynak Eksenine Göre Doğrultusu	**İlgili Metal**	**ϕ ve Ω**	**Karakteristik Gerilme (F_{nBM} veya F_{nw})**	**Etkin Alan (A_{BM} veya A_{we})**	**Kaynak Metali Gerekli Dayanım Düzeyi**
Kesme	Esas Metal	**Bölüm 13.4**			**Bölüm 13.2.6**
	Kaynak Metali	$\phi = 0.75$ $\Omega = 2.00$	$0.60F_E$	**Bölüm 13.2.2.1**	
Kaynak eksenine paralel çekme ve basınç	Kaynak eksenine paralel çekme ve basınç etkilerinin kaynak tasarımında göz önüne alınmasına gerek yoktur.				
DAİRESEL VE OVAL DOLGU KAYNAKLAR					
Birleşim yüzeyine paralel etkin alanda kesme	Esas Metal	**Bölüm 13.4**			**Bölüm 13.2.6**
	Kaynak Metali	$\phi = 0.75$ $\Omega = 2.00$	$0.60F_E$	**Bölüm 13.2.3.1**	

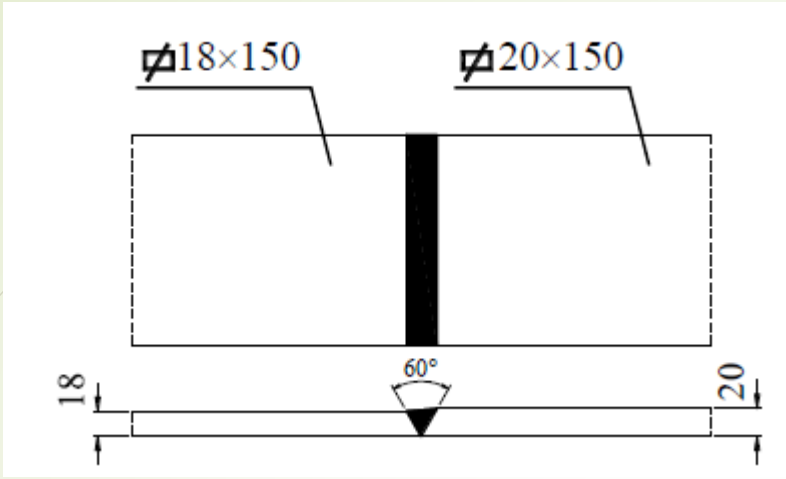
Örnek

- Aşağıdaki şekilde detayı verilen Levha 18x150 ve Levha 20x150 enkesitli elemanların küt kaynaklı ek birleşimi sabit ve hareketli yükler altında sırasıyla, *PG* = 100kN ve *PQ* = 250kN eksenel çekme kuvveti etkisindedir. Kaynak uygulamasında, kaynağın başlatıldığı ve sonlandırıldığı yardımcı levhalar (başlangıç ve bitiş levhaları) kullanılmıştır. Kaynak malzemesi E70 elektrodu ve çelik malzemesi olarak S275 alınız.

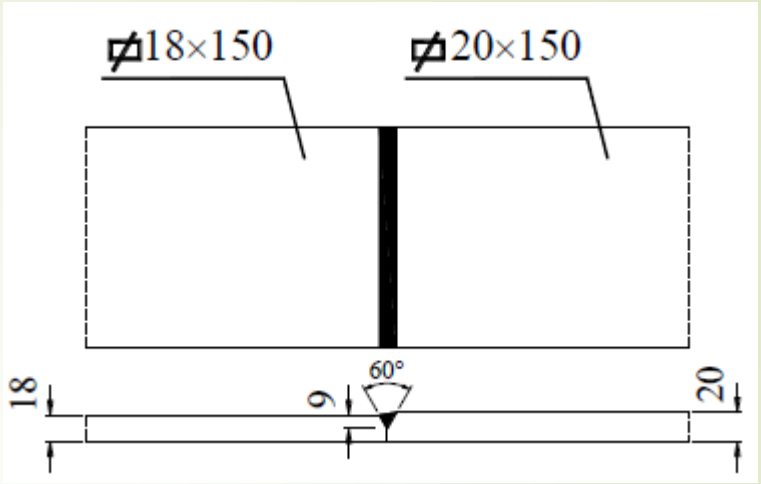
a. Tam penetrasyonlu küt kaynak kullanılması veya

b. Kaynak ağzı şekli V-60° ve derinliği 9mm olan kısmi penetrasyonlu küt kaynak kullanılması durumunda

birleşimin karakteristik dayanımının belirlenmesi ve tasarım (YDKT) dayanımlarının kontrolü.



Tam penetrasyonlu küt kaynak uygulaması



Kısmi penetrasyonlu küt kaynak uygulaması

- S 275 F_y = 27,5 kN/cm^2 F_U = 43,0 kN/cm^2 (*Yönetmelik* Tablo 2.1A)

Tam penetrasyonlu küt kaynak kullanılması durumunda

Tam penetrasyonlu küt kaynağın etkin kalınlığı, *a* (*Yönetmelik* 13.2.1.1)

a = 18 mm

- Tam penetrasyonlu küt kaynaklı birleşimin karakteristik dayanımı, *Rnw*

(*Yönetmelik* Tablo 13.5 Kaynak eksenine dik çekme) Esas metal dayanımı belirleyicidir.

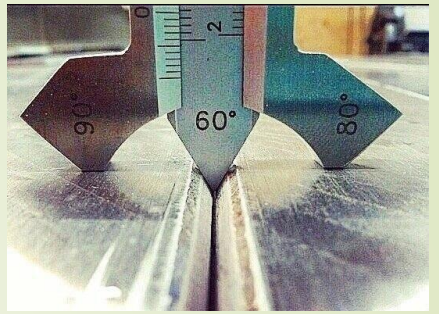
- Rn=FyAg
- Rn=27,5*1,8*15=742,5 kN

$P_u=1,2P_g+1,6P_q$

$P_u=1,2*100+1,6*250=520$ kN

$R_d=0,90*R_n=0,90*742,5=668,25$

$R_d>P_u$ OK $P_u/R_d=520/668=0,78<1,0$



- Kısmi penetrasyonlu küt kaynak kullanılması durumunda

Kısmi penetrasyonlu küt kaynak işlemi (*Yönetmelik* Tablo 13.1)

Örtülü elektrot ile elektrik ark kaynağı, V - 60° düz kaynak.

Kısmi penetrasyonlu küt kaynağın etkin kalınlığı, *a* (*Yönetmelik* Tablo 13.1)

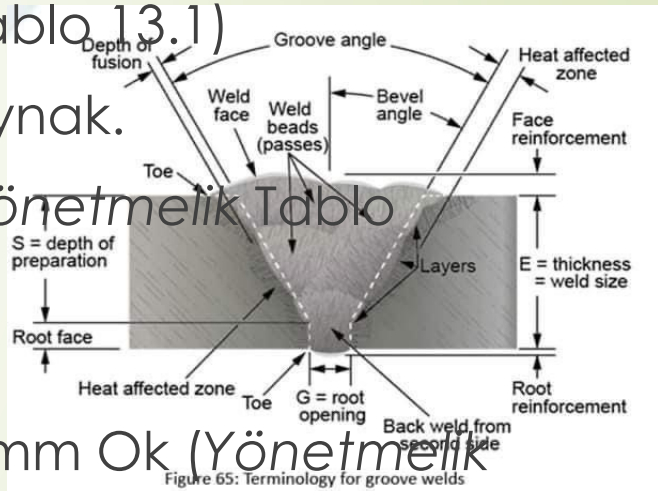


Figure 65: Terminology for groove welds

- *a* = 9 mm (Kaynak ağzı derinliği)
- 19 mm ≥ t = 18 mm > 13 mm a_{min} = 6 mm < *a* = 9 mm Ok (*Yönetmelik* Tablo 13.3)

Kısmi penetrasyonlu küt kaynaklı birleşimin karakteristik dayanımı

- Kaynak metali karakteristik çekme kuvveti dayanımı, R_{nw}

(*Yönetmelik* Tablo 13.5 Kaynak eksenine dik çekme)

- $R_{nw}=F_{nw}A_{we}$
- $F_{nw}=0,60F_E=0,60*48=28,8$ kN/cm^2
- $A_{we}=0,9*15=13,5$ cm^2 (*Yönetmelik* 13.2.1.1)
- $R_{nw}=28,8*13,5=388,8$ kN

Esas metal karakteristik çekme kuvveti dayanımı, R_{nBM}

(*Yönetmelik* Tablo 13.5 Kaynak eksenine dik çekme)

- $R_{nBM}=F_{nBM}A_{BM}$
- $F_{nBM}=F_u$ $A_{BM}=A_e$
- $R_{nBM}=43,0*0,9*15=580,5$ kN

$P_u=1,2P_g+1,6P_q$

$P_u=1,2*100+1,6*250=520$ kN

$R_d=0,80*R_{nw}=0,80*388,8=311,04$

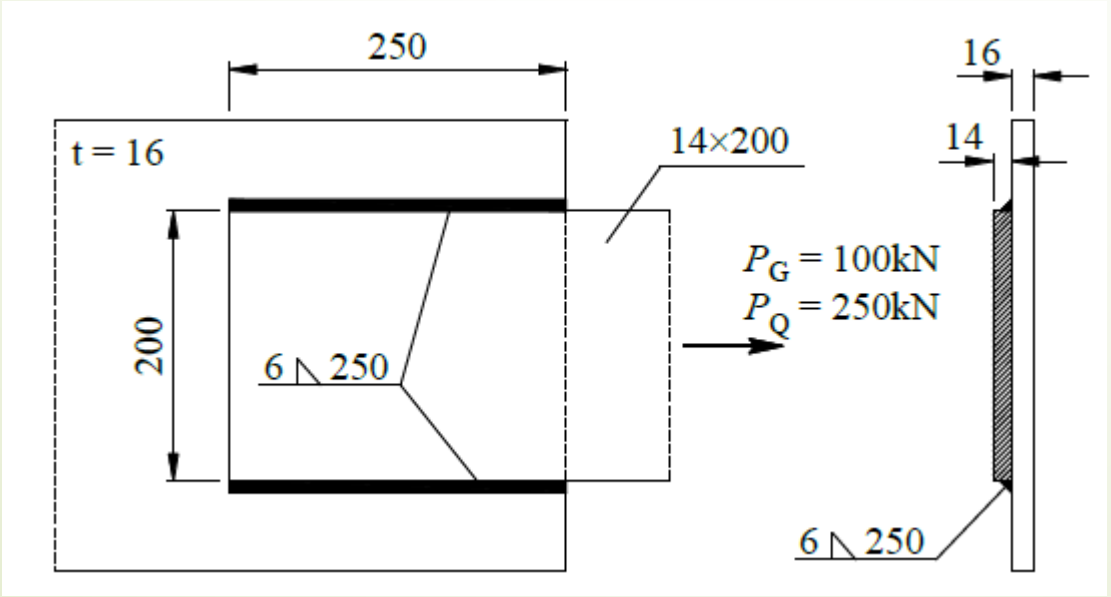
$R_d=0,75*R_{nBM}=0,75*580,5=435,38$

$R_d<P_u$ $P_u/R_d=520/311,04=1,67>1,0$ **not safe**

Örnek 13.3

Aşağıdaki şekilde köşe kaynaklı uç birleşim detayı verilen Levha 14□200 enkesit boyutlarına sahip eleman, sabit ve hareketli yükler altında sırasıyla, P_G = 100kN ve P_Q = 250kN eksenel çekme kuvveti etkisindedir. Çelik malzemesi S355 Kaynak malzemesini seçiniz.

- Birleşimin karakteristik dayanımının belirlenmesi
- Birleşimin tasarım dayanımının kontrolü (YDKT)



- *FE* = 550 N/mm2 > *Fu* = 510 N/mm2 (*Yönetmelik* 2.4)
- Birleşimin karakteristik dayanımının belirlenmesi
- *Yönetmelik* 13.2.2 uyarınca, birleşen eleman kenarı ile birleşilen yüzey arasındaki 90° lik açının uygunluğu
- 60° < 90° < 120°

Köşe kaynağın etkin kalınlığı, *a* (13.2.2.2)

- 13 mm < t = 14 mm ≤ 19 mm *a*min = 4 mm < *a* = 6 mm OK (*Yönetmelik* Tablo 13.4)
- amaks=0,7*(14,0-2,0)=8,4 mm >a=6,0mm
- Köşe kaynağın etkin uzunluğu, *Le* (*Yönetmelik* 13.2.2.2)
- L=250-2a=250-2*6,0=238 mm (*Yönetmelik* 13.2.2)
- *L* = 238 mm ≥ Maks.(6 * 6,0 = 36 mm; 40 mm) = 40 mm OK

- L = 238 mm ≥ b = 200 mm
- L = 238 mm ≤ 150a = 150(6.0) = 900 mm olduğundan L_e = 238 mm

Köşe kaynağın etkin alanı, A_{we} (*Yönetmelik* 13.2.2.1)

- Awe = $\sum aL_e$
- Awe = 2*6,0*238 = 2856 mm2

Kaynak metali karakteristik dayanımı, *R*nw (*Yönetmelik* Tablo 13.5 Kesme)

- Rnw = *Fnw a* (Birim kaynak uzunluğu için) Fnw =0.60*FE* , *a* = 6.0mm
- Rnw = *Fnw Awe* (Toplam kaynak alanı için)
- Fnw = 0,60*55 = 33,0 kN/cm2
- Rnw =33 *0,6 = 19,80 kN/cm
- Rnw = 2*19,80*23,8 = 942,48 kN

Esas metal karakteristik dayanımı, R_{nBM} (*Yönetmelik* 13.4.2)

Kayma etkisinde akma ve kırılma sınır durumu için

- $R_{nBM} = F_{nBM}\, t$ $\qquad F_{nBM} = 0{,}60F_u$
- R_{nBM} = 0,60 * 51,0 * 1,4 = 42,84 kN/cm , R_{nBM} = 0,60 * 35,5 * 1,4 = 29,82 kN/cm
- R_{nBM} = 2*42,84*23,8 = 2039,2 kN , R_{nBM} = 2*29,82*23,8 = 1419,3 kN

Birleşimin karakteristik dayanımının belirlenmesinde kaynak metali dayanımı belirleyicidir.

- Rn = Min(Rnw ; RnBM) = Min (0,75*942,48 ; 0,75*2039,2 ; 1,0*1419,3)
- Rn = 942.48 kN , Rd=0,75*942,48=706,86 kN

$Pu=1,2Pg+1,6Pq$

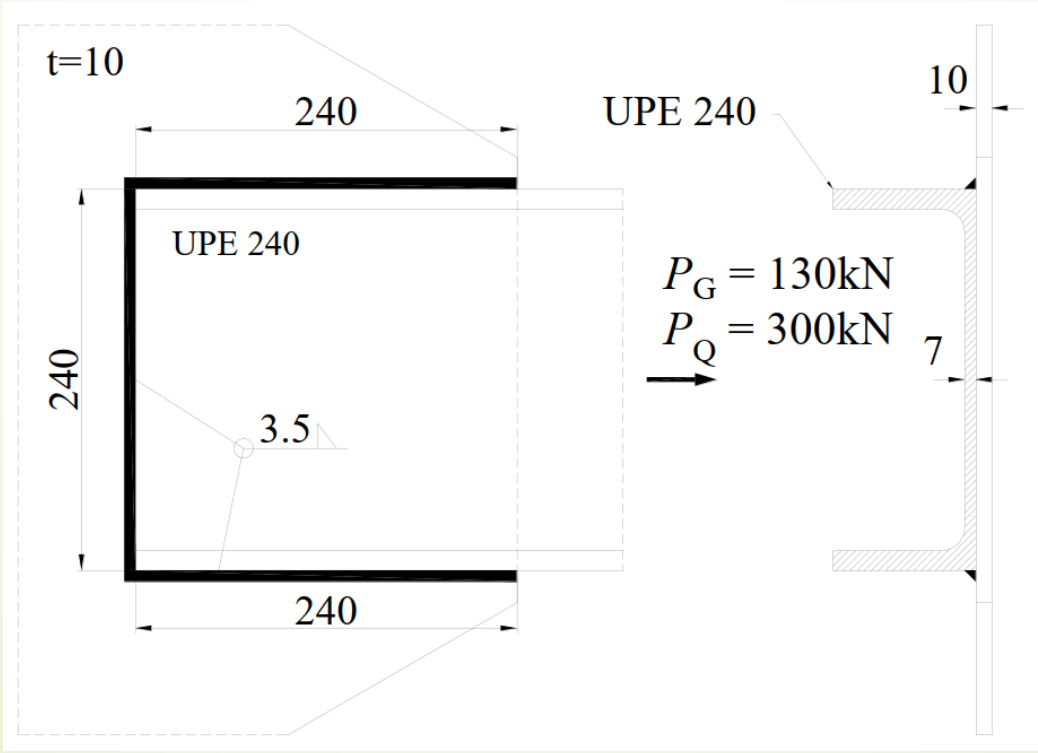
$Pu=1,2*100+1,6*250=520\ kN$

$Rd=0,75Rnw=0,75*942,48=706,86$

$Rd>Pu \qquad Pu/Rd=520/706 = 0,74<1,0$

Örnek 13.3

- Aşağıdaki şekilde köşe kaynaklı uç birleşim detayı verilen UPE 240 enkesitli eleman, sabit ve hareketli yükler altında sırasıyla, *PG* = 130kN ve *PQ* = 300kN eksenel çekme kuvveti etkisindedir. Çelik malzemesi S275
- **a.** Birleşimin karakteristik dayanımının belirlenmesi
- **b.** Birleşimin tasarım dayanımının kontrolü (YDKT)



- *FE* = 550 N/mm2 > *Fu* = 430 N/mm2 (*Yönetmelik* 2.4)
- Birleşimin karakteristik dayanımının belirlenmesi
- *Yönetmelik* 13.2.2 uyarınca, birleşen eleman kenarı ile birleşilen yüzey arasındaki 90° lik açının uygunluğu
- 60° < 90° < 120°

Köşe kaynağın etkin kalınlığı, *a* (13.2.2.2) **(Alın kaynağı için)**

- 6 mm < t = 7 mm ≤ 13 mm *a*min = 3,5 mm < *a* = 3,5 mm OK (*Yönetmelik* Tablo 13.4)
- amaks=0,7*(7,0-2,0)=3,5 mm >a=3,5 mm
- Köşe kaynağın etkin uzunluğu, *Le* (*Yönetmelik* 13.2.2.2)
- L=2450-a=240-3,5=236,5 mm (*Yönetmelik* 13.2.2)
- *L* = 236,5 mm ≥ Maks.(6 * 6,0 = 36 mm; 40 mm) = 40 mm OK

- L = 236,5 mm ≤ 150a = 150(3,5) = 525 mm olduğundan L_e = 236,5 mm

- Kaynak metali karakteristik dayanımı, *R*nw (*Yönetmelik* 13.2.4(b))
  Yönetmelik 13.2.4(b) uyarınca, üniform kalınlıklı kaynak grubunun karakteristik dayanımı,
  Denk.(13.5) ve Denk.(13.6) ile hesaplanan değerlerin **büyüğüne eşit** alınacaktır.

- $R_{nw} = R_{nwl} + R_{nwt}$
- $R_{nw} = 0{,}85R_{nwl} + 1{,}5R_{nwt}$
- $R_{nwl} = F_{nw} * A_{we} = 0{,}6 * F_E * A_{we} = 0{,}6 * 55{,}0 * 2 * 0{,}35 * 23{,}6 = 546{,}32\ kN$

- $R_{nwt} = F_{nw} * A_{we} = 0{,}6 * F_E * A_{we} = 0{,}6 * 55{,}0 * 0{,}35 * 24{,}0 = 277{,}20\ kN$
- $R_{nw} = R_{nwl} + R_{nwt} = 546{,}32 + 277{,}20 = 823{,}52\ kN$

- $R_{nw} = 0{,}85R_{nwl} + 1{,}5R_{nwt} = 0{,}85 * 546{,}32 + 1{,}5 * 277{,}20 = 880{,}17\ kN$
- $R_{nw} = 880{,}17\ kN$

$Pu=1,2Pg+1,6Pq$

$Pu=1,2*130+1,6*300=636\ kN$

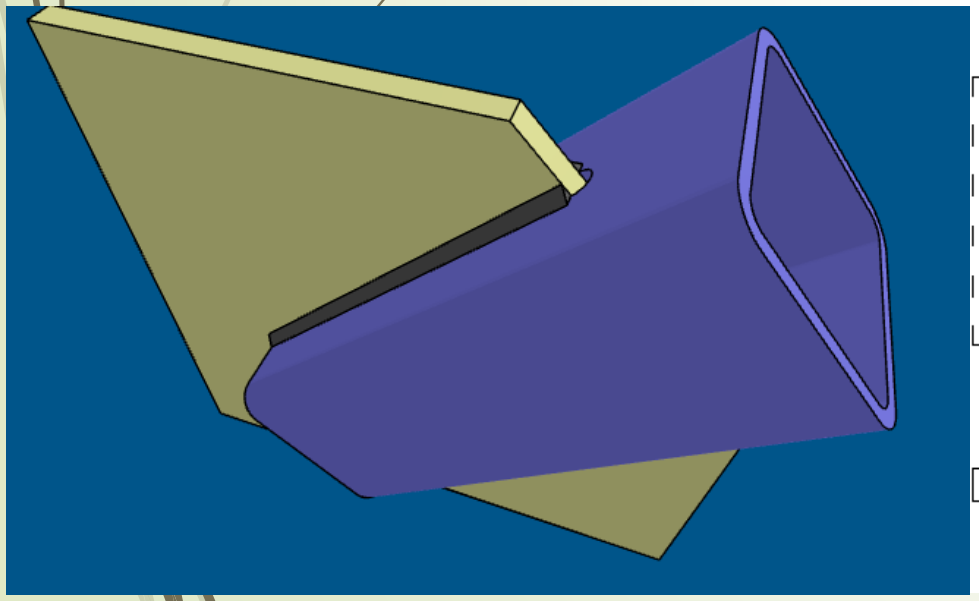
$Rd=0,75Rnw=0,75*880,17=660,13$

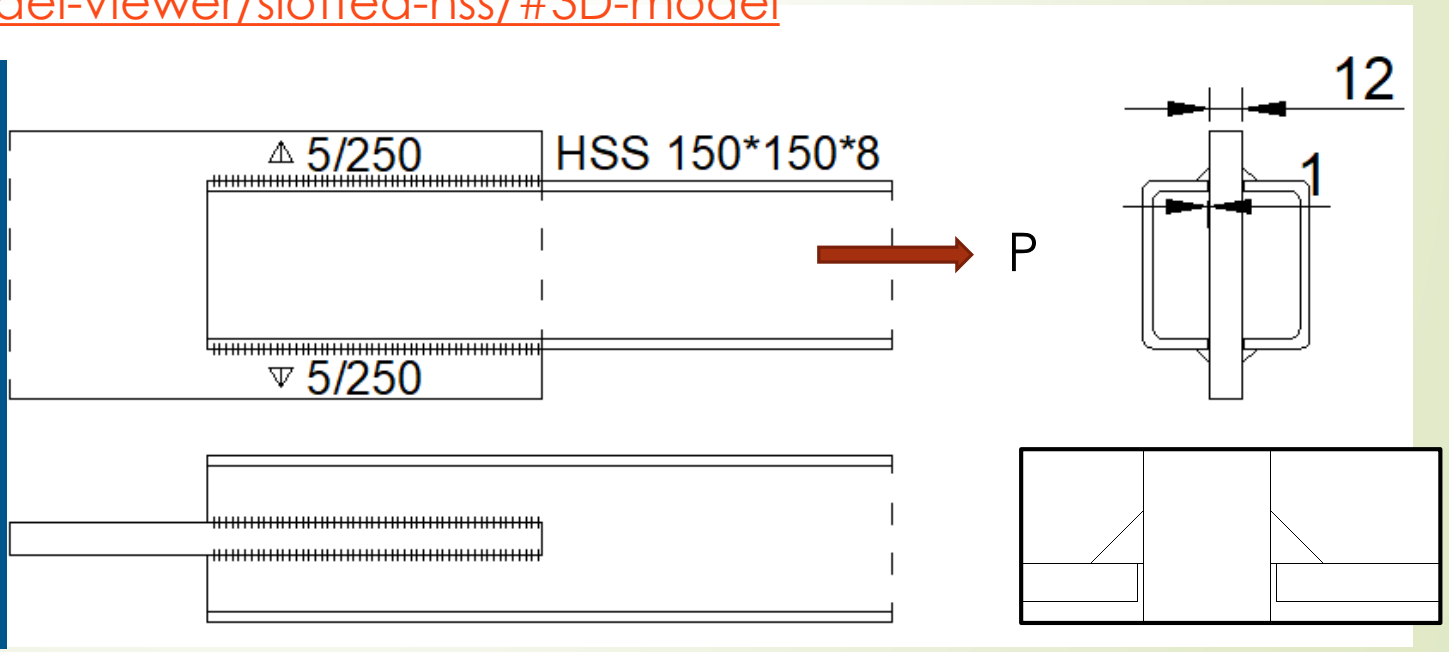
$Rd>Pu \qquad Pu/Rd=636/660,13 = 0,96<1,0$

Örnek

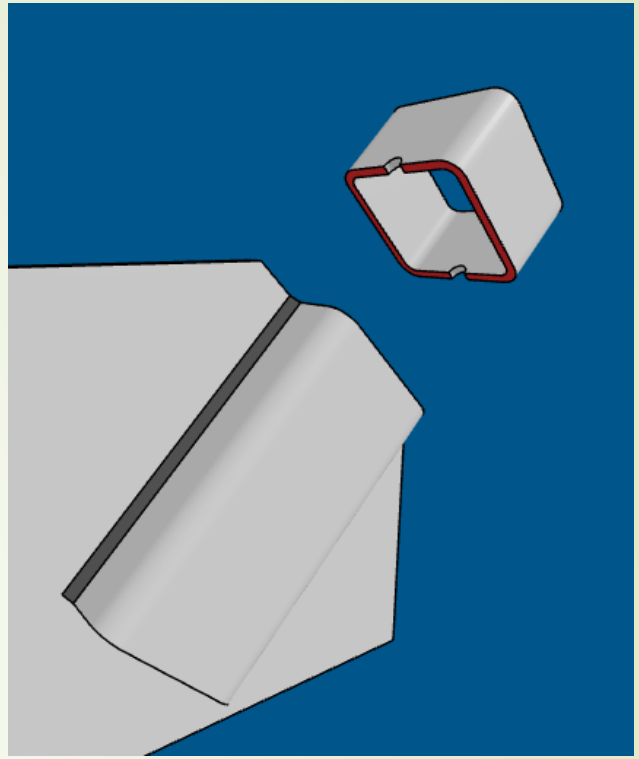
- A square hollow structural section (150*150*8) is used as a tension member and is welded to gusset plate as shown in the figure. Compute the design strength of the connection. S355 steel is used for all members. Do not consider weld strength. In order to facilitate construction of the hollow structural section on gusset plate, the HSS section is cut by additional 2 mm more than the thickness of the gusset plate.

https://aisc-org.github.io/aisc-model-viewer/slotted-hss/#3D-model





Çekme çubuğu hesabı



Limit durumlar

- Akma sınır durumu için

Tn=Fy*Ag=35,5*43,2=1533,6 kN

ϕ*Tn=0,90*1533,6=1380 kN

- Kırılma sınır durumu için

Tn=Fu*Ae Ae=U*An L=250 mm H=150 mm

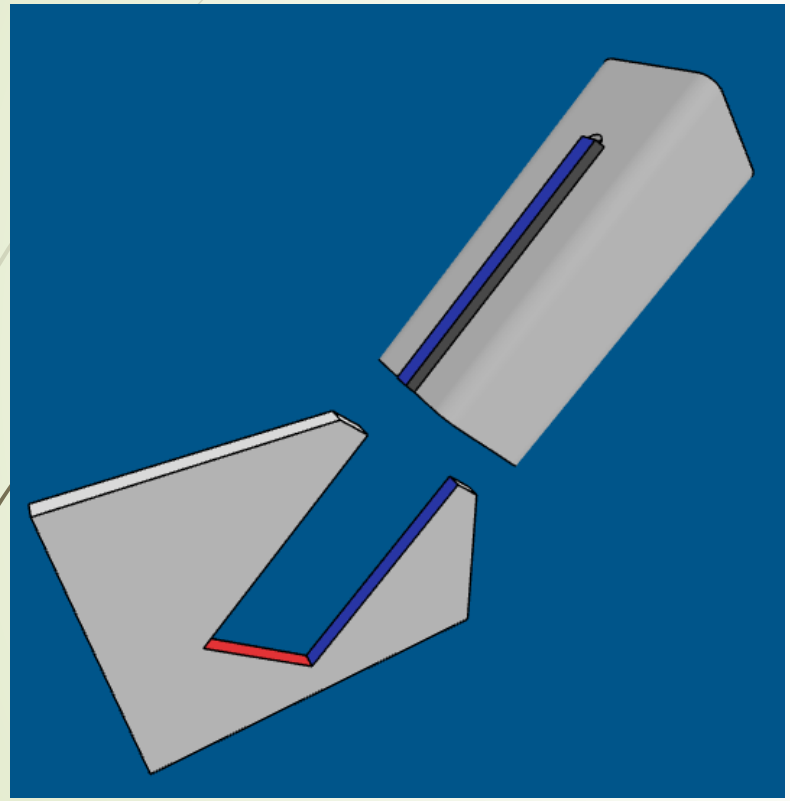
$$\overline{x} = \frac{B^2+2*B*H}{4*(B+H)} = \frac{150^2+2*150*150}{4*(150+150)} = 56{,}25\ mm \Rightarrow U = 1 - \frac{\overline{X}}{L} = 1 - \frac{56{,}25}{250} = 0{,}775$$

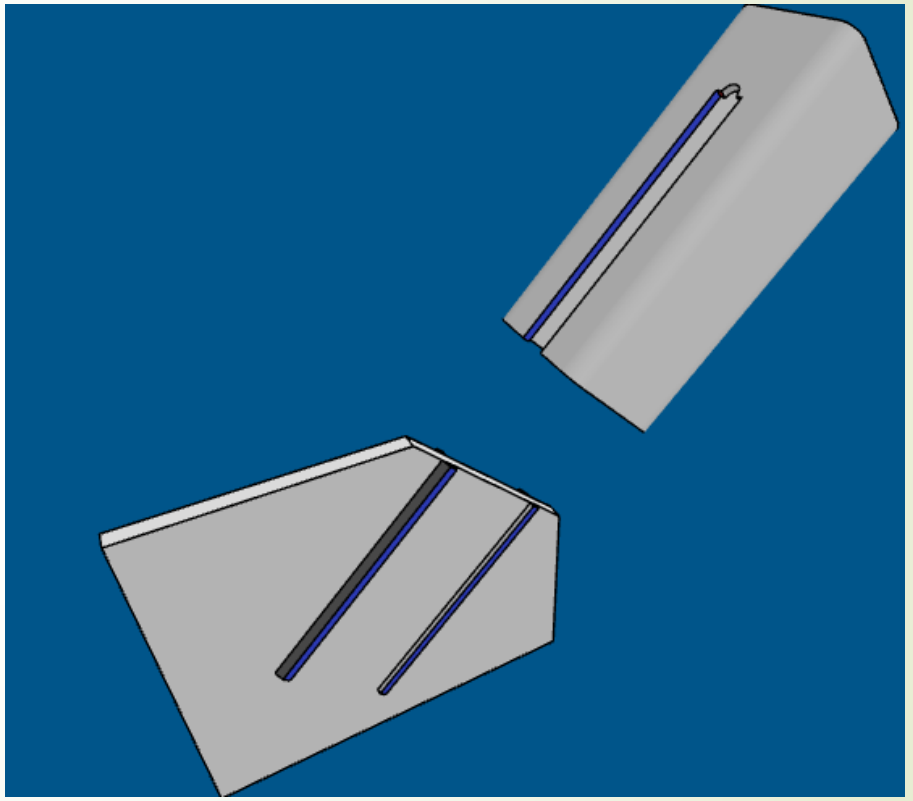
An=43,2-2*(1,2+0,2)*0,8=40,96 cm2 Ae=0,775*40,96=31,74 cm2

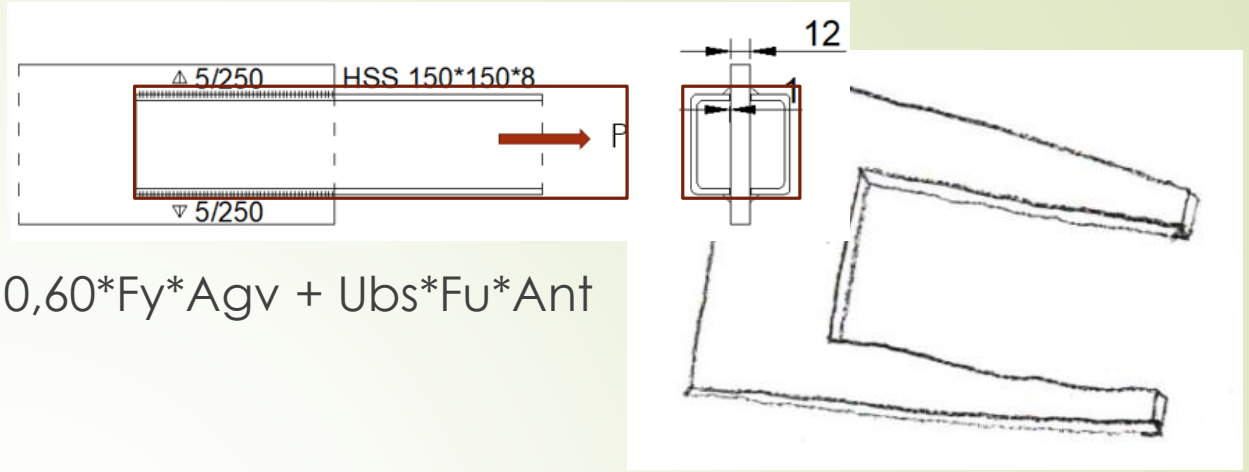
Tn=Fu*Ae=51,0*31,74=1618,7 kN

ϕ*Tn=0,75*1618,7=1214 kN

Blok Shear







- Blok kesme sınır durumu için
- Durum 1

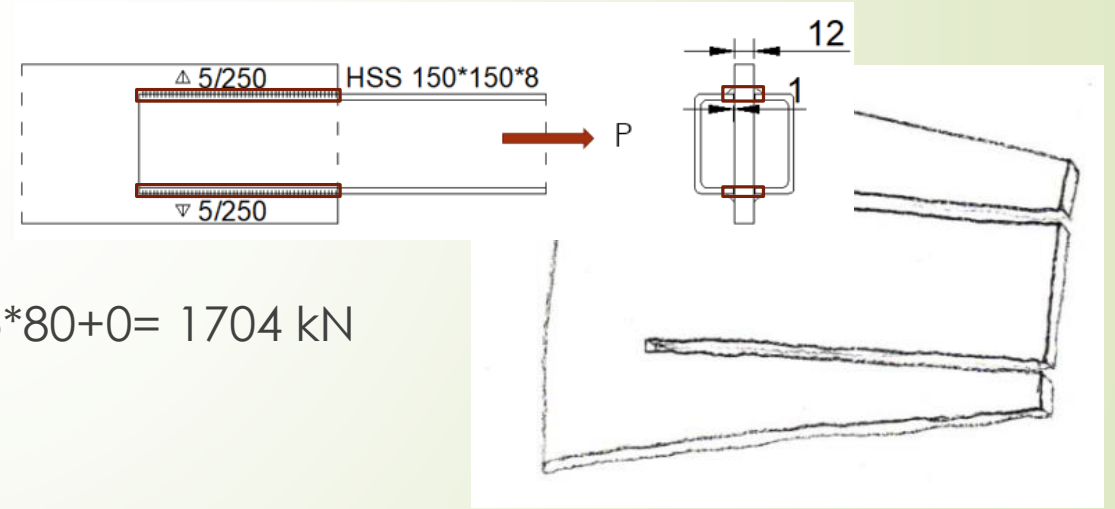
Rn=0,60*Fu*Anv + Ubs*Fu*Ant ≤ 0,60*Fy*Agv + Ubs*Fu*Ant

Agv=Anv=2*25,0*1,2=60 cm2

Ant=15*1,2=18 cm2

Rn=0,6*51,0*60+1*51,0*18=2754 kN ≤ 0,60*35,5*60+1*51,0*18= 2196 kN

Rn=2196 kN ϕ*Rn=0,75*2196=1647 kN



- Durum 2

Agv=4*25,0*0,8=80 cm2

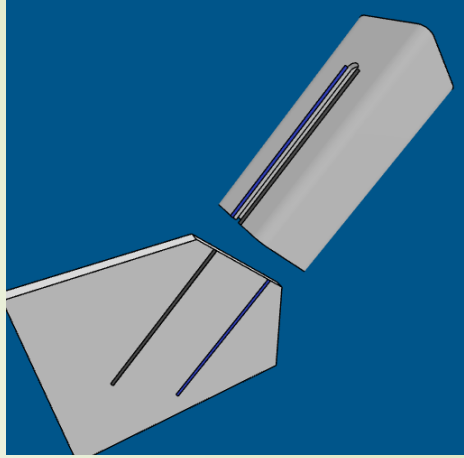
Ant~0

Rn=0,6*51,0*80+0=2448 kN ≤ 0,60*35,5*80+0= 1704 kN

Rn=1704 kN ϕ*Rn=0,75*1704=1278 kN

- Çekme Kapasitesi
- Min(1380 , 1214 , 1647 , 1278) = 1214 kN Kırılma sınır durumu etkindir.

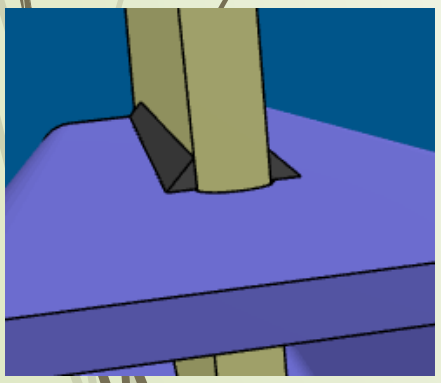
Kaynak hesabı

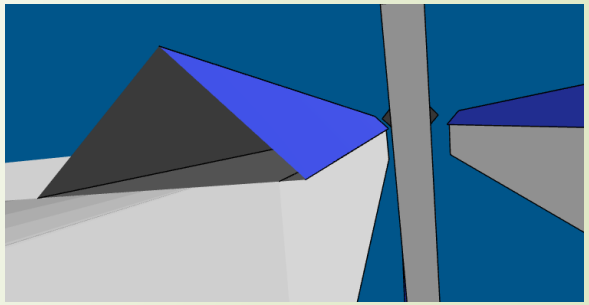


- *FE* = 550 N/mm2 > *Fu* = 510 N/mm2 (*Yönetmelik* 2.4)
- Birleşimin karakteristik dayanımının belirlenmesi
- *Yönetmelik* 13.2.2 uyarınca, birleşen eleman kenarı ile birleşilen yüzey arasındaki 90° lik açının uygunluğu
- 60° < 90° < 120°

Köşe kaynağın etkin kalınlığı, *a* (13.2.2.2)

- 6 mm < t = 8 mm ≤ 13 mm *a*min = 3,5 mm < *a* = 5 mm OK (*Yönetmelik* Tablo 13.4)





- a_{maks}=1,0*(8,0)=8,0 mm > a=5,0mm
- Köşe kaynağın etkin uzunluğu, L_e (*Yönetmelik* 13.2.2.2)
- L=250-2a=250-2*5,0=240 mm (*Yönetmelik* 13.2.2)
- *L* = 240 mm ≥ Maks.(6 * 5,0 = 30 mm; 40 mm; 50 mm) = 50 mm OK

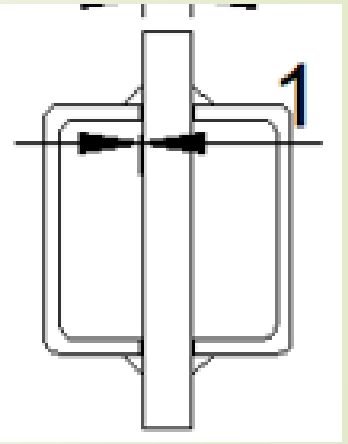
- L = 240 mm ≥ b = 150 mm
- L = 240 mm ≤ 150a = 150(5.0) = 750 mm olduğundan L_e = 240 mm

Köşe kaynağın etkin alanı, A_{we} (*Yönetmelik* 13.2.2.1)

- $A_{we} = \sum a^* L_e$
- A_{we} = 4*5,0*240 = 4800 mm2

Kaynak metali karakteristik dayanımı, R_{nw} (*Yönetmelik* Tablo 13.5 Kesme)

- $R_{nw} = F_{nw}^* a$ (Birim kaynak uzunluğu için) F_{nw} =0.60*FE a = 5.0mm
- $R_{nw} = F_{nw}^* A_{we}$ (Toplam kaynak alanı için)
- F_{nw} = 0,60*55 = 33,0 kN/cm2
- R_{nw} =33 *0,5 = 16,50 kN/cm
- R_{nw} = 4*16,50*24,0 = 1584 kN → Rnw=33,0*48,0=1584 kN



Esas metal karakteristik dayanımı, R_{nBM} (*Yönetmelik* 13.4.2)

Kayma etkisinde kırılma sınır durumu için

- $R_{nBM} = F_{nBM}$ t F_{nBM} = 0,60F_u , F_{nBM} = 0,60F_y
- R_{nBM} = 0,60 * 51,0 * 1,2 = 36,72 kN/cm , R_{nBM} = 0,60 * 35,5 * 1,2 = 25,56 kN/cm
- R_{nBM} = 2*36,72*24,0 = 1763 kN , R_{nBM} = 2*25,56*24,0 = 1226,9 kN
- Base plate t=12 mm HSS t=8*2=16 mm olduğundan HSS duvarı gusset plate göre daha kalın olduğundan HSS kesitte kesme oluşmaz.

Birleşimin karakteristik dayanımının belirlenmesinde kaynak metali dayanımı belirleyicidir.

- $R_n = Min(R_{nw} ; R_{nBM}) = Min (\mathbf{1584*0,75} ; 1763*0,75 ; 1226,9*1,0)$
- $R_n = 1584*0,75 = 1188$ kN

Profilden gelen yükle karşılaştırma

- Çekme kapasitesinden P_u=1214 kN

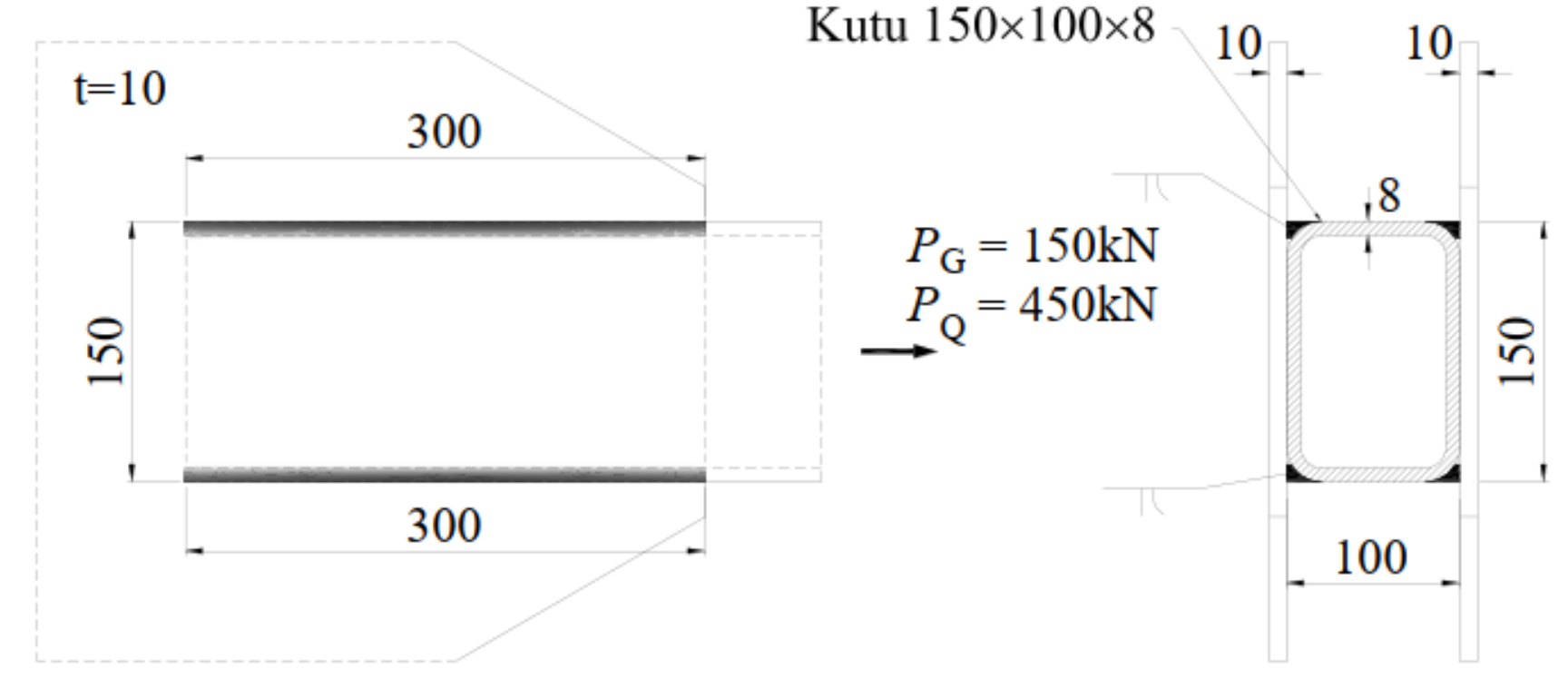
Kaynak kapasitesi Rd=1188 kN

$R_d<P_u$ $P_u/R_d=1214/1188 = 1,022>1,0$ Uygun değildir

- $A_{we} = 4*5,5*240 = 5280$ mm^2 Kaynak kalınlığı artırılırsa (önceki 4800 mm^2)
- $F_{nw} = 0,60*55 = 33,0$ kN/cm^2
- $R_{nw} = 33*0,55 = 18,15$ kN/cm
- $R_{nw} = 4*18,15*24,0 = 1742$ kN
- $R_n = Min(R_{nw} ; R_{nBM}) = Min (1742*0,75 ; 1763*0,75 ; 1226,9*1,0)$
- $R_n = 1226,9$ kN
- $R_d>P_u$ $P_u/R_d=1214/1226,9 = 0,99<1,0$ Uygundur.

Örnek 13.4

- Aşağıdaki şekilde verilen ve örtülü elektrot ile elektrik ark kaynağı işlemi uygulanarak teşkil edilen uç birleşim detayına sahip Kutu 150×100×8 enkesitli eleman, sabit ve hareketli yükler altında sırasıyla, *PG* = 150kN ve *PQ* = 450kN eksenel çekme kuvveti etkisindedir. Çelik malzemesi S275
  a. Birleşimin karakteristik dayanımının belirlenmesi
  b. Birleşimin tasarım dayanımının kontrolü (YDKT)



- Enkesitinin bir kenarı eğrisel küt kaynak (*Yönetmelik* 13.2.1.1, Şekil 13.2)
  *F*E = 550 N/mm2 > *F*u = 430 N/mm2 (*Yönetmelik* 2.4)

a. Birleşimin karakteristik dayanımının belirlenmesi Enkesitinin bir kenarı eğrisel küt kaynağın etkin kalınlığı, *a* (*Yönetmelik* Tablo 13.2) Kaynak işlemi örtülü elektrot ile elektrik ark kaynağı olup kaynak metali eğrisel yüzey hizasına kadar doldurulmaktadır.

- $a = 5/16\,R, \quad R = 2t = 2 * 8{,}0 = 16\,mm$ *Yönetmelik* Tablo 13.2)
- $a = 5/16 * 16 = 5{,}0\,mm$ Enkesitinin bir kenarı eğrisel küt kaynağın etkin uzunluğu, *L*e (*Yönetmelik* 13.2.2.2)
  L = 300-2*a* =300-2*5,0=290 mm (*Yönetmelik* 13.2.2)
- *L* = 290 mm ≥ Maks.(6 × 6.0 = 36 mm; 40 mm) = 40 mm ✓
  L = 290 mm ≥ *b* = 150 mm ✓
  L = 290 mm ≤ 150*a* = 150(6.0) = 900 mm olduğundan *L*e = 290 mm
  Kaynak metali karakteristik dayanımı, *R*nw
  (*Yönetmelik* Tablo 13.5 Kesme)
- $R_{nw} = F_{nw} * A_{we} = 0{,}6 * F_E * A_{we} = 0{,}6 * 55{,}0 * 4 * 0{,}5 * 29{,}0 = 1914\,kN$Esas metal karakteristik dayanımı, *R*nBM (*Yönetmelik* 13.4.2)

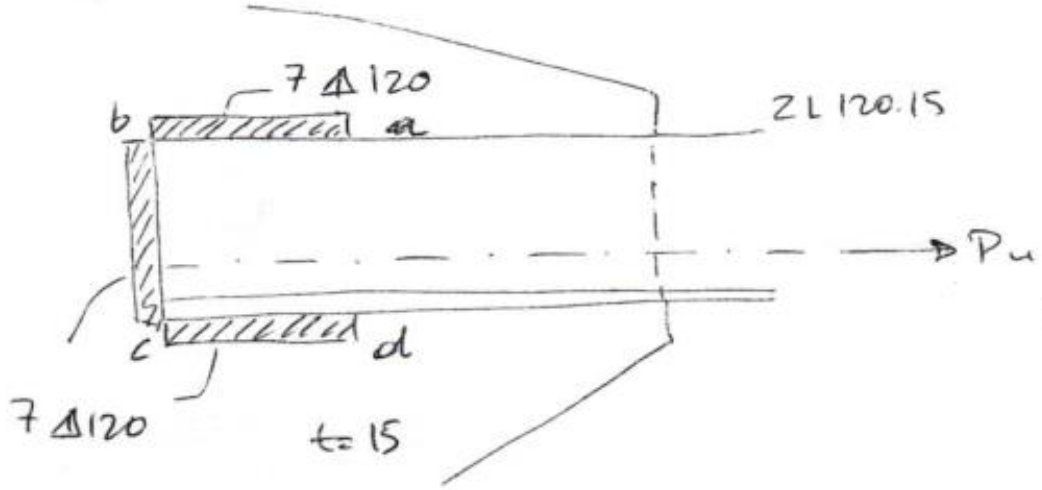
Kayma etkisinde kırılma sınır durumu için

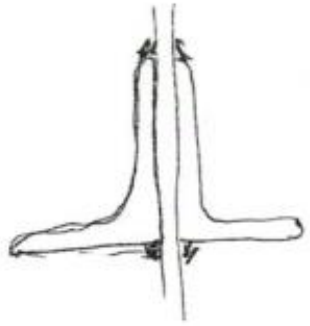
- $R_{nBM} = F_{nBM} * a$ (Birim kaynak uzunluğu için)
- $R_{nBM} = F_{nBM} * A_{BM}$ (Toplam kaynak alanı için)
- $F_{nBM} = 0{,}6F_u, \quad F_{nBM} = 0{,}6F_y$
- $R_{nBM} = 0{,}6 * 43{,}0 * 0{,}8 = 20{,}64\ kN/cm$ $\quad R_{nBM} = 0{,}6 * 27{,}5 * 0{,}8 = 13{,}20\ kN/cm$
- $R_{nBM} = 2 * 0{,}6 * 43{,}0 * 0{,}8 * 29{,}0 = 2394{,}24\ kN$
- $R_{nBM} = 2 * 0{,}6 * 27{,}5 * 0{,}8 * 29{,}0 = 1531{,}0\ kN$
- Gerekli çekme kuvveti dayanımı
- P_U=1,2*P_g+1,6*P_q =1,2*150+1,6*450=900 kN
- Birleşimin tasarım dayanımının kontrolü
- $R_d = \emptyset_w * R_{nw} = 0{,}80 * 1914{,}0 = 1531{,}0\ kN$
- $R_d = \emptyset_w * R_{nBM} = 0{,}75 * 2394{,}24 = 1795{,}68\ kN$
- $R_d = \emptyset_w * R_{nBM} = 1{,}0 * 1531{,}0 = 1531{,}0\ kN$
- $\frac{R_u}{R_d} = \frac{900}{1531{,}0} = 0{,}59 < 1{,}0\ uygundur$

Question 1

(1)

A connection detail which is under pure tension effect given in figure. $P_D = 300$; $P_L = 200$ kN. Steel material is S355.





(1 ksi = 6,89 MPa)

S355

$F_y = 355$ MPa

$F_u = 550$ MPa

E70

$F_u = 70$ ksi

$\cong 480$ MPa

L 120.15

$e_x = e_y = 3.51$ cm

$A_g = 33,9$ cm²

Note: welds at a-b, b-c and c-d are applied seperately.

a) Check the maximum stress in connection. (Use a conservative approach)

b) Please re-arrange weld lengths so that makes this connection a simple connection.

Thickness check

$t = 12$ $\quad 13 \leq t \leq 19 \longrightarrow a_{min} = 4.0$ mm

$t = 12$ $\quad 0{,}7\,(t - 0{,}2) = 7$ mm $\quad a_{max} = 7$ mm ✓

Length check

$d = \frac{120 - 2.7}{} > (6.7 = 42$ mm, 40 mm$)$

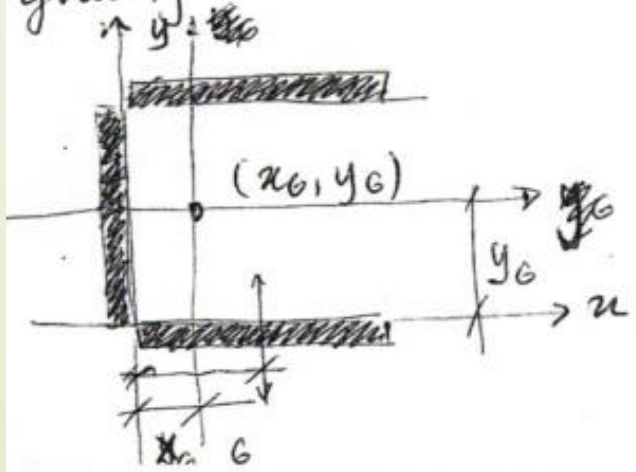
$L = \frac{120 - 2.7}{} < 150.7 = 1050$ mm

$\rightarrow L_e = L$

$L_e = 120 - 2.7 = 106$ mm

$A_{e_1} = 0{,}7 \cdot (10{,}6) = 7.42$ cm^2

- Since the connection is eccentric, gravity center should be located.



$y_G = 6$ cm

$$x_G = \frac{A_{e1} \cdot 6 + A_{e1} \cdot 6 + A_{e1} \cdot 0}{3 A_{e1}} = 4 \text{ cm}$$

$$I_x = \frac{0,7.(10,6)^3}{12} + 2.(7,42.6^2) = 603,72 \ cm^4$$

$$I_y = 2.\left(\frac{0,7.(10,6)^3}{12} + 7,42.2^2\right) + 7,42.4^2 = 317,03 \ cm^4$$

$I_o \cong J \cong I_x + I_y = 603,72 + 317,03 = 920,75 \ cm^4$ → Polar moment of inertia

$P_u = 1,2.P_D + 1,6 P_L = 1,2.300 + 1,6.200 = 680 \ kN$

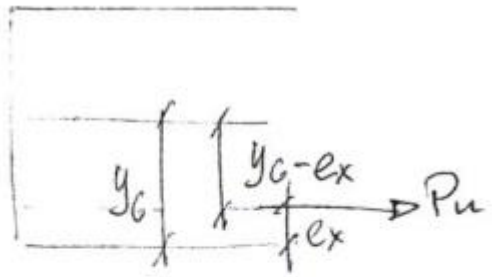
→ Calculation for one-side of the connection, the applied force can be divided by half.

$P_u = \frac{680}{2} = 340 \ kN$ (for each side)

→ moment due to eccentricity.

$e = y_G = e_x = 6 - 3,51 \ cm = 2,49$; $M_u = e.P_u = 2,49.340 = 846,60 \ kNcm$

$f_p = \frac{P}{A_{weld}}$; $f_{px} = \frac{P_x}{A_{weld}}$; $f_{py} = \frac{P_y}{A_{weld}}$



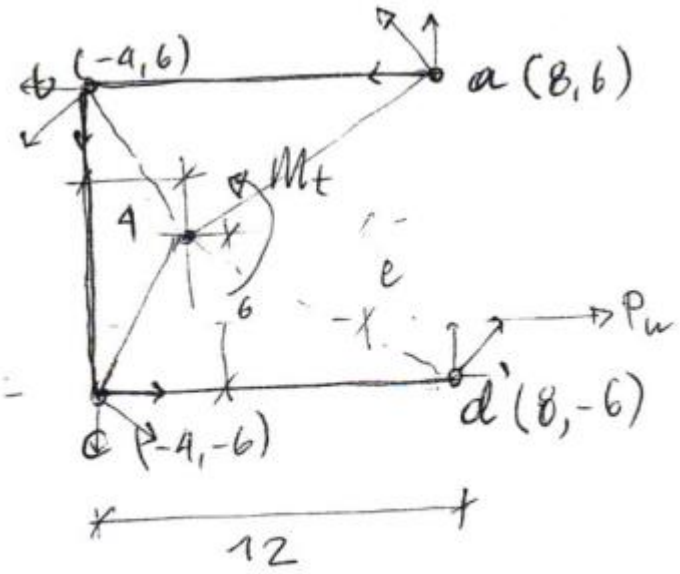
$$f_p = \frac{P}{A_{weld}} \; ; \; f_{px} = \frac{P_x}{A_{weld}} \; ; \; f_{py} = \frac{P_y}{A_{weld}}$$

f_m (torsional moment) $= \frac{M_t}{J} \cdot r$; $f_{mx} = \frac{M_t}{J}|y|$; $f_{my} = \frac{M_t}{J}|x|$

x, y: abs. distances

Stress due to tension $\longrightarrow f_{px} = \frac{340}{3.7,12} = 15,27 \text{ kN/cm}^2$; $f_{py} = 0$

Stress due to torsional moment

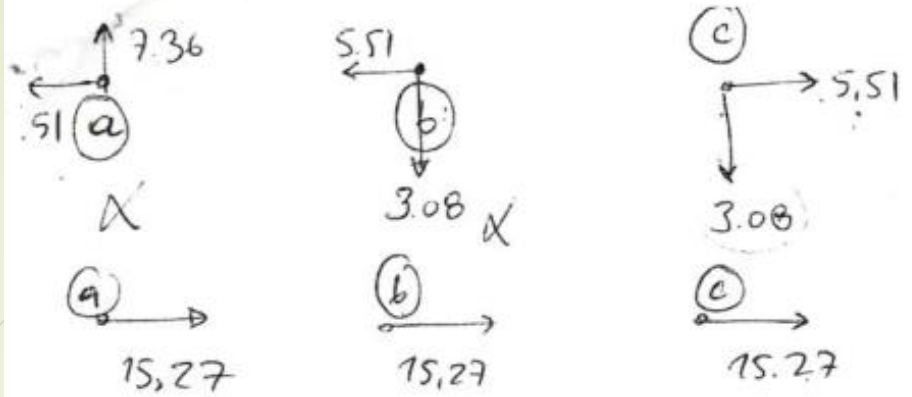


(a) $f_{mx}^a = \frac{846,6}{920,75} \cdot 6 = 5,51 \text{ kN/cm}^2$ $\quad f_{my}^a = \frac{846,6}{920.75} \cdot 8 = 7.36 \text{ kN/cm}^2$

(b) $f_{mx}^b = \frac{846,6}{920,75} \cdot 6 = 5,51$ $\quad f_{my}^b = \frac{846.6}{920.75} \cdot |-4| = 3,08 \text{ kN/cm}^2$

(c) $f_{mx}^c = \frac{846.6}{920.75} |-6| = 5,51$ $\quad f_{my}^c = \frac{846.6}{920,75} \cdot |-4| = 3.08 \text{ kN/cm}^2$

(d) $f_{mx}^d = \frac{846.6}{920.75} |-6| = 5.51$ $\quad f_{my}^d = \frac{846,6}{920.75} \cdot 8 = 7.36 \text{ kN/cm}^2$



f_{mx}, f_{my} (3)

f_{px}, ~~f_{py}~~

D point

$f_x = 15,27 + 5,51 = 20.88$ kN/cm²

$f_y = 0 + 7,36 = 7.36$ kN/cm²

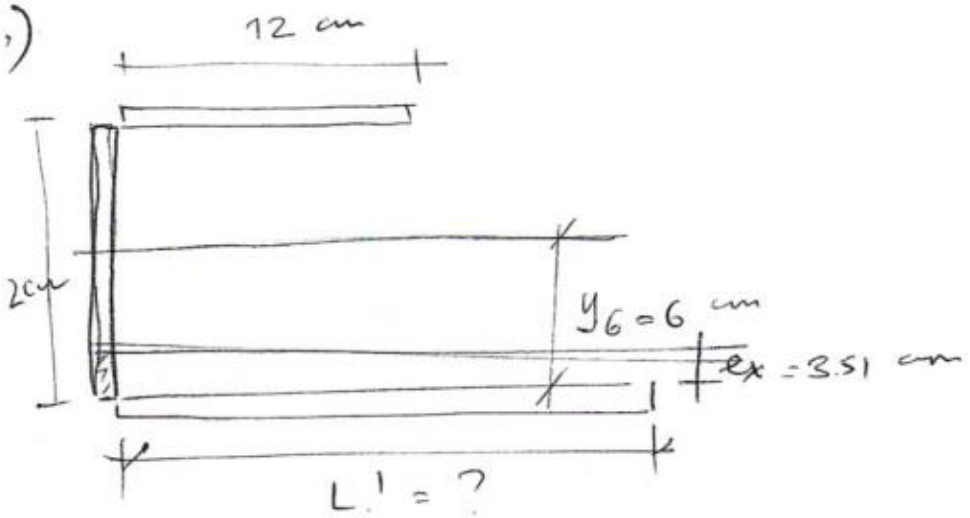
$f_R = \sqrt{f_x^2 + f_y^2} = 22,14$ kN/cm²

$F_E = 70$ ksi $\simeq 48,0$ kN/cm²

$F_{nw} = 0,6 \cdot 48 = 28$ kN/cm²

$\phi F_{nw} = 0.75 \cdot 28 = 21$ kN/cm² $< f_R = 22,14$ kN/cm² (Not safe)

;)



$$(L' \cdot a) \cdot e_x + (a \cdot e_x)\frac{e_x}{2} = ((12 - e_x) \cdot a)\frac{12 - e_x}{2} + (12 \cdot a)(12 - e_x)$$

$$(0{,}7 \cdot L') \cdot 3{,}51 + (0{,}7 \cdot 3{,}51) \cdot \frac{3{,}51}{2} = ((12 - 3{,}51) \cdot 0{,}7)\left(\frac{12 - 3{,}51}{2}\right) + (12 \cdot 0{,}7)(12 - 3{,}51)$$

$$2{,}457 \cdot L' + 4{,}312 = 25{,}228 + 71{,}316$$

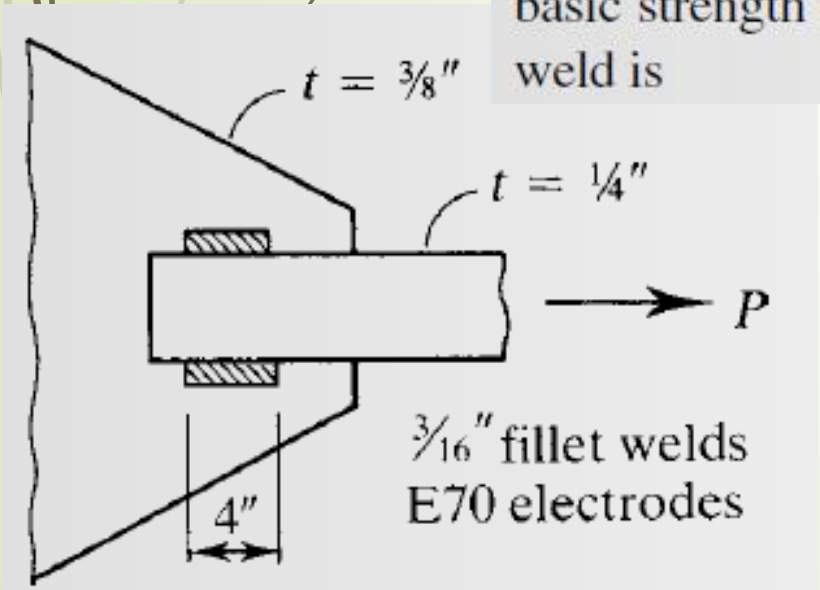
$L' = 37{,}53$ cm

If bottom weld length would be applied as determined L'; connection would not have eccentricity thus no moment.

Örnek

A plate used as a tension member is connected to a gusset plate, as shown in Figure 7.40. The welds are 3⁄16-inch fillet welds made with E70XX electrodes. The connected parts are of A36 steel. Assume that the tensile strength of the member is adequate, and determine the available strength of the welded connection.

Because the welds are placed symmetrically about the axis of the member, this connection qualifies as a simple connection, and there is no additional load due to eccentricity. Since both weld segments are parallel to the applied load, $\theta = 0°$ and the basic strength of the weld is $F_{nw} = 0.60F_{EXX}$. The nominal load capacity per inch of weld is



$$R_n = 0.707wF_{nw} = 0.707\left(\frac{3}{16}\right)(0.6 \times 70) = 5.568 \text{ kips/in.}$$

The design strength of the weld is

$$\phi R_n = 0.75(5.568) = 4.176 \text{ kips/in.}$$

Check the base-metal shear. Since both components are of the same type of steel, the smaller thickness will control. The shear yield strength is

$$\phi R_n = \phi(0.6F_y t) = 1.00(0.6)(36)\left(\frac{1}{4}\right) = 5.4 \text{ kips/in.}$$

The shear rupture strength is

$$\phi R_n = \phi(0.6F_u t) = 0.75(0.6)(58)\left(\frac{1}{4}\right) = 6.525 \text{ kips/in.}$$

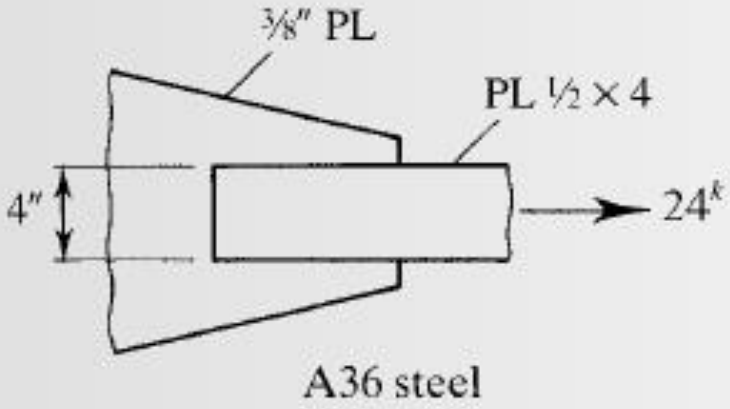
The base-metal shear strength is therefore 5.4 kips/in., and the weld shear strength controls. For the connection,

$$\phi R_n = 4.176 \text{ kips/in.} \times (4 + 4) \text{ in.} = 33.4 \text{ kips}$$

Design strength of the weld is 33.4 kips.

Örnek

A plate ½ × 4 of A36 steel is used as a tension member to carry a service dead load of 6 kips and a service live load of 18 kips. It is to be attached to a ⅜-inch gusset plate, as shown in Figure 7.44. Design a welded connection.



The base metal in this connection is A36 steel, so E70XX electrodes will be used. No restrictions on connection length have been stipulated, so weld length will not be limited, and the smallest permissible size will be used:

$$\text{Minimum size} = \frac{3}{16} \text{ in.} \qquad \text{(AISC Table J2.4)}$$

Try a 3/16-in. fillet weld, using E70XX electrodes. The design strength per inch is

$$1.392D = 1.392(4) = 4.176 \text{ kips/in.}$$

The shear yield strength of the base metal is

$$0.6F_y t = 0.6(36)\left(\frac{3}{8}\right) = 8.1 \text{ kips/in.}$$

and the shear rupture strength of the base metal is

$$0.45F_u t = 0.45(58)\left(\frac{3}{8}\right) = 9.788 \text{ kips/in.}$$

The weld strength of 4.176 kips/in. governs. The factored load is

$$P_u = 1.2D + 1.6L = 1.2(6) + 1.6(18) = 36 \text{ kips}$$

and

$$\text{Required length} = \frac{36}{4.176} = 8.62 \text{ in.}$$

$$\text{Minimum length} = 4w = 4\left(\frac{3}{16}\right) = 0.75 \text{ in.} < 8.62 \text{ in.} \qquad \text{(OK)}$$

Use two 4.5-in.-long side welds for a total length of $2 \times 4.5 = 9$ in. For this type of connection, the length of the side welds must be at least equal to the transverse distance between them, or 4 inches in this case. The provided length of 4.5 inches will therefore be adequate.

Use a 3/16-inch fillet weld with E70XX electrodes, with a total length of 9 inches, as shown in Figure 7.45.

Örnek

Design a welded connection. The given loads are service loads. Use $F_y = 50$ ksi for the angle tension member and $F_y = 36$ ksi for the gusset plate. Show your results on a sketch, complete with dimensions.

a. Use LRFD.

b. Use ASD.

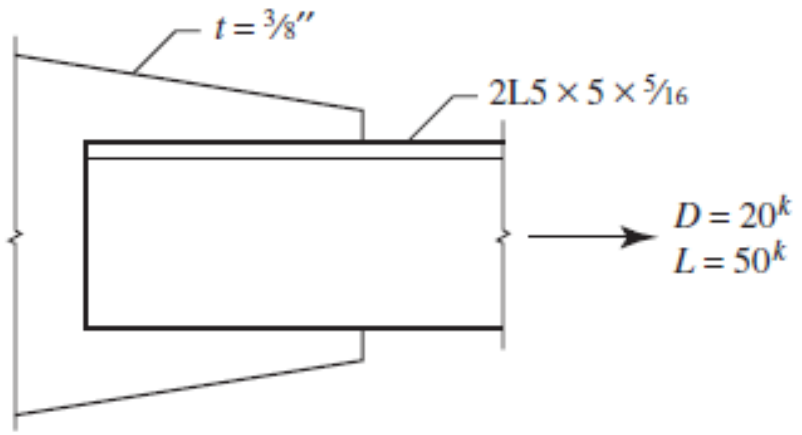


FIGURE P7.11-5

7.11-5

From AISC Table J2.4, the minimum weld size is 3/16 inch (based on the angle thickness). Maximum size = 5/16 – 1/16 = 1/4 in.

(a) LRFD solution: $P_u = 1.2D + 1.6L = 1.2(20) + 1.6(50) = 104.0$ kips

Try w = ¼ in., $\phi R_n = 1.392 \times 4$ sixteenths = 5.568 kips/in.

The base metal shear yield strength (gusset plate controls) is

$$0.6F_y t = 0.6(36)\left(\frac{3}{8}\right) = 8.1 \text{ kips/in.}$$

Shear rupture strength is $0.45F_u t = 0.45(58)\left(\frac{3}{8}\right) = 9.788$ kips/in.

Base metal shear yield strength for the angles is

$$0.6F_y t = 0.6(50)\left(\frac{5}{16} \times 2\right) = 18.75 \text{ kips/in.}$$

and the shear rupture strength is $0.45F_u t = 0.45(65)\left(\frac{3}{8} \times 2\right) = 21.94$ kips/in.

The weld strength of 5.568 kips/in. governs. Both longitudinal and transverse welds will be used. To determine the required length of the longitudinal welds, investigate the two options specified in AISC J2.4(c). First, assuming the same strength for both the longitudinal and transverse welds,

$$\text{total required length of weld} = \frac{104}{5.568} = 18.68 \text{ in.}$$

$$\text{length of longitudinal welds} = \frac{18.68 - 5}{2} = 6.84 \text{ in.}$$

For the second option, the strength of the longitudinal welds is

$$0.85(5.568) = 4.733 \text{ kips/in.}$$

and the strength of the transverse weld is

$$1.5(5.568) = 8.352 \text{ kips/in.}$$

The load to be carried by the longitudinal welds is

$$104 - 5(8.352) = 62.24 \text{ kips}$$

so the required length of the longitudinal welds is

$$\frac{62.24}{2(4.733)} = 6.575 \text{ in.}$$

The first option requires shorter longitudinal welds. Try a 5-inch transverse weld and two 7-inch longitudinal welds. Check the block shear strength of the gusset plate.

$$A_{gv} = A_{nv} = 2 \times \frac{3}{8}(7) = 5.25 \text{ in.}^2 \qquad A_{nt} = \frac{3}{8}(5) = 1.875 \text{ in.}^2$$

$$R_n = 0.6F_uA_{nv} + U_{bs}F_uA_{nt} = 0.6(58)(5.25) + 1.0(58)(1.875) = 291.5 \text{ kips}$$

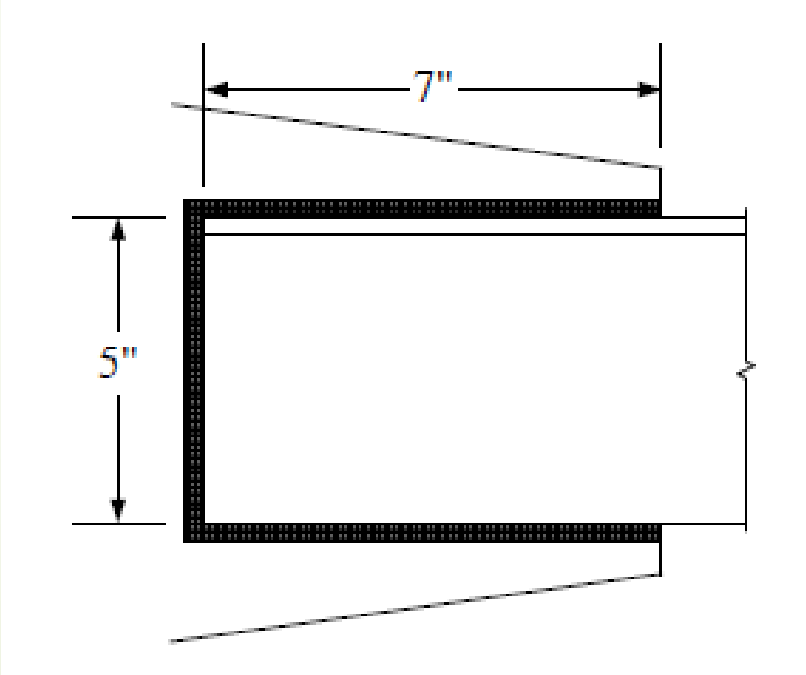
with an upper limit of

$$0.6F_yA_{gv} + U_{bs}F_uA_{nt} = 0.6(36)(5.25) + 1.0(58)(1.875)$$

The design strength is

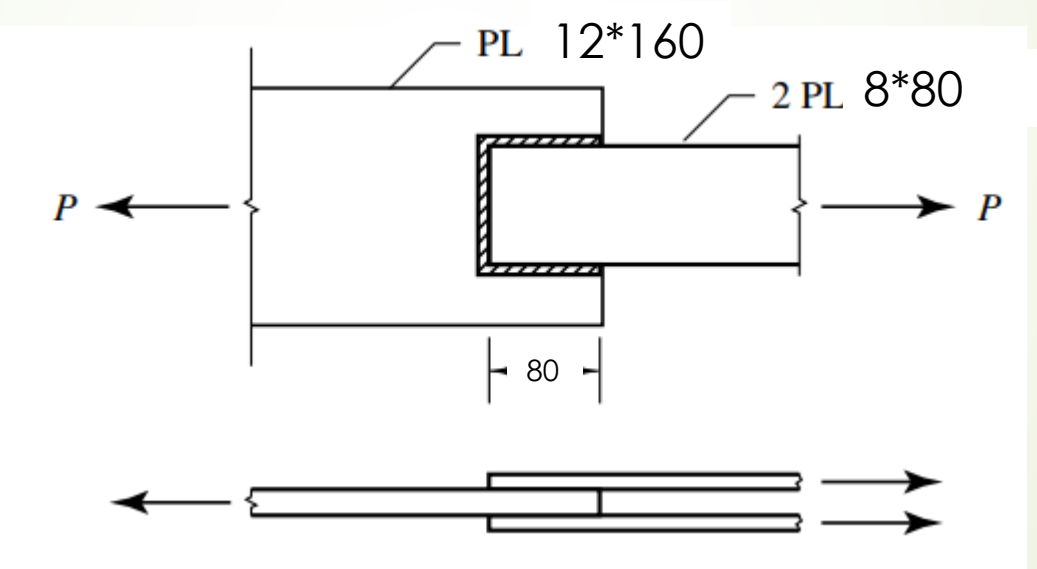
$$\phi R_n = 0.75(222.2) = 167 \text{ kips} > 104 \text{ kips} \qquad \text{(OK)}$$

Use ¼-in. fillet welds as shown.



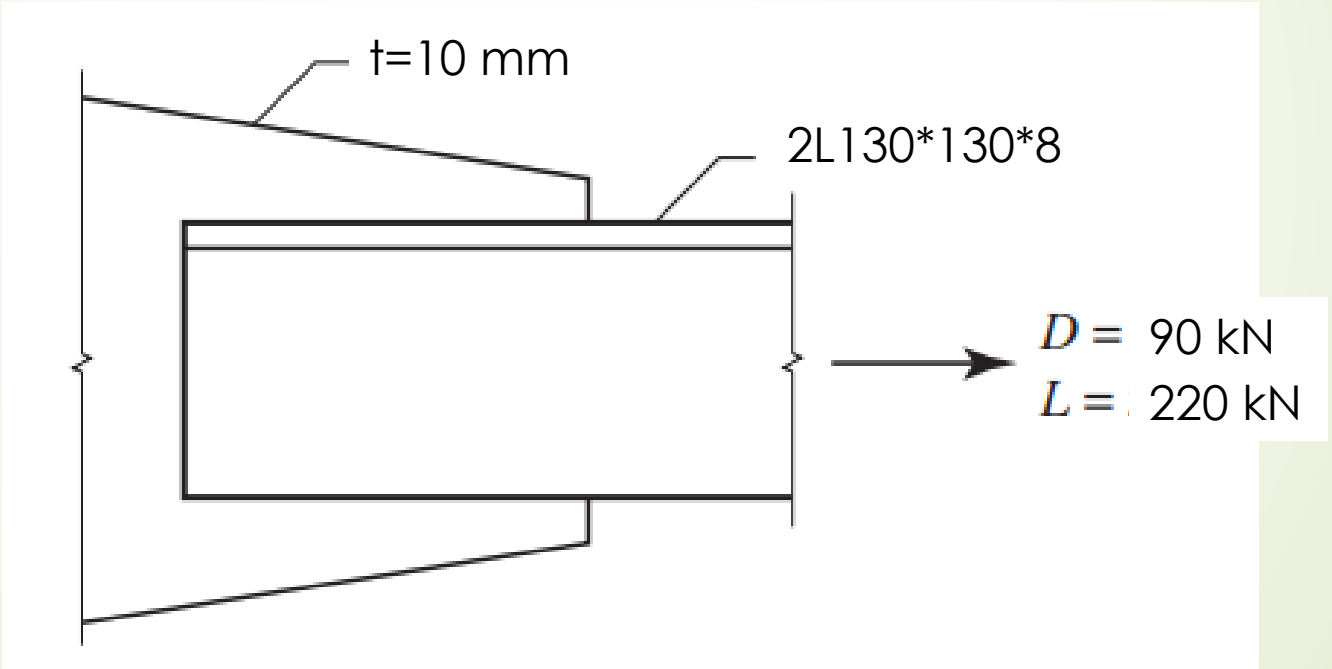
Working Problems

- A tension member splice is made with 4,0 mm E70 fillet welds as shown in Figure. Each side of the splice is welded as shown. The inner member is a PL 12 × 80 and each outer member is a PL 8×80. All steel is S235. Determine the maximum service load, *P*, that can be applied if the live load to dead load ratio is 3.0.



Working Problems

- Design a welded connection. The given loads are service loads. Use *Fy* = 355 MPa for the angle tension member and *Fy* = 235 MPa for the gusset plate. Show your results on a sketch, complete with dimensions.



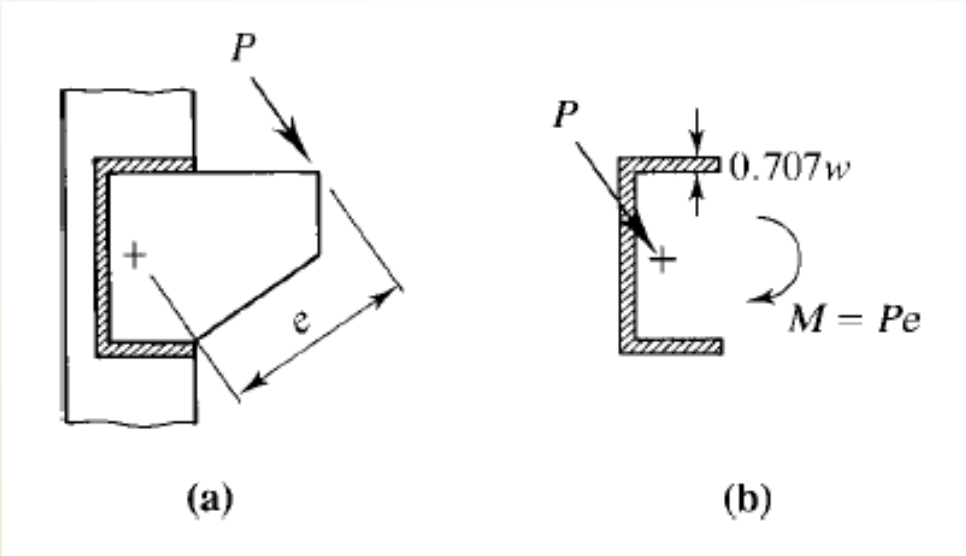
Ekzantrik yüklenmiş kaynaklı birleşimler

- Elastik analizle dayanım hesabı
- Maksimum yükle dayanım hesabı

Elastik analizle dayanım hesabı

- Kolaylık olması açısından kaynak kalınlıkları (0.707*w) hesaba katılır
- Birleşimdeki yükler x ve y doğrultusunda hesap edilerek toplam kuvvet hesap edilir.

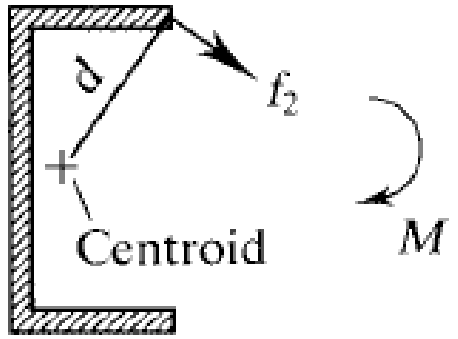
kesme kuvveti için $f_{1x} = \frac{P_x}{L}$ ve $f_{1y} = \frac{P_y}{L}$ ve moment ten gelen $f_2 = \frac{Md}{J}$



Moment için

$$f_2 = \frac{Md}{J}$$

- X ve y doğrultularında ayrı ayrı hesap edilirse
- $f_{2x} = \frac{My}{J}$ ve $f_{2y} = \frac{Mx}{J}$
- M Moment değeri
- X ve y ağırlık merkezine uzak olan mesafeler
- J polar atalet momenti

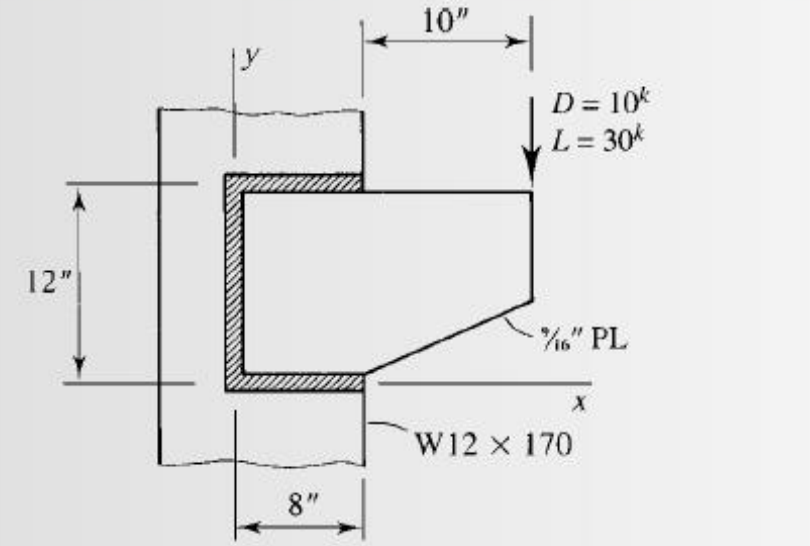


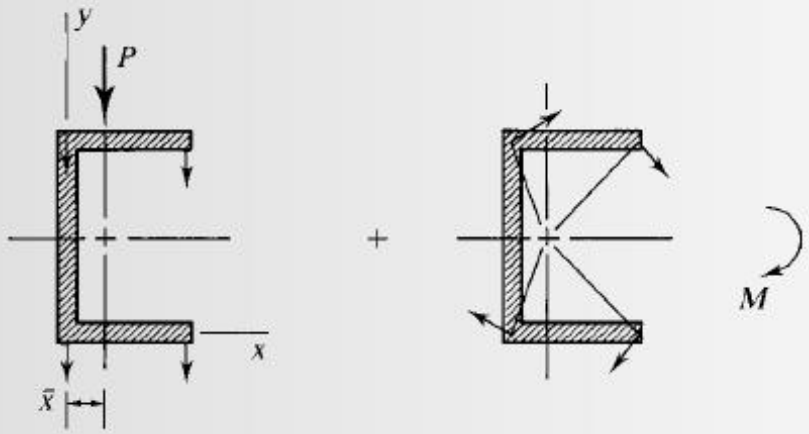
$$J = \int_A r^2 dA = \int_A (x^2 + y^2) dA = \int_A x^2 \, dA + \int_A y^2 \, dA = I_y + I_x$$

- Toplam kuvvet $f_v = \sqrt{(\Sigma f_x)^2 + (\Sigma f_y)^2}$

Örnek

Determine the size of weld required for the bracket connection in Figure 8.18. The service dead load is 10 kips, and the service live load is 30 kips. A36 steel is used for the bracket, and A992 steel is used for the column.





$$P_u = 1.2D + 1.6L = 1.2(10) + 1.6(30) = 60 \text{ kips}$$

The eccentric load may be replaced by a concentric load and a couple, as shown in Figure 8.18. The direct shearing stress is the same for all segments of the weld and is equal to

$$f_{1y} = \frac{60}{8+12+8} = \frac{60}{28} = 2.143 \text{ ksi}$$

Before computing the torsional component of shearing stress, the location of the centroid of the weld shear area must be determined. From the principle of moments with summation of moments about the y axis,

$$\bar{x}(28) = 8(4)(2) \qquad \text{or} \qquad \bar{x} = 2.286 \text{ in.}$$

The eccentricity e is $10 + 8 - 2.286 = 15.71$ in., and the torsional moment is

$$M = Pe = 60(15.71) = 942.6 \text{ in.-kips}$$

If the moment of inertia of each horizontal weld about its own centroidal axis is neglected, the moment of inertia of the total weld area about its horizontal centroidal axis is

$$I_x = \frac{1}{12}(1)(12)^3 + 2(8)(6)^2 = 720.0 \text{ in.}^4$$

Similarly,

$$I_y = 2\left[\frac{1}{12}(1)(8)^3 + 8(4-2.286)^2\right] + 12(2.286)^2 = 195.0 \text{ in.}^4$$

and

$$J = I_x + I_y = 720.0 + 195.0 = 915.0 \text{ in.}^4$$

Figure 8.18 shows the directions of both components of stress at each corner of the connection. By inspection, either the upper right-hand corner or the lower right-hand corner may be taken as the critical location. If the lower right-hand corner is selected,

$$f_{2x} = \frac{My}{J} = \frac{942.6(6)}{915.0} = 6.181 \text{ ksi}$$

$$f_{2y} = \frac{Mx}{J} = \frac{942.6(8 - 2.286)}{915.0} = 5.886 \text{ ksi}$$

$$f_v = \sqrt{(6.181)^2 + (2.143 + 5.886)^2} = 10.13 \text{ ksi} = 10.13 \text{ kips/in.}$$

for a unit throat dimension.

Check the strength of the base metal. The bracket is the thinner of the connected parts and controls. From Equation 7.35, the base-metal shear *yield* strength per unit length is

$$\phi R_n = 0.6F_y t = 0.6(36)\left(\frac{9}{16}\right) = 12.2 \text{ kips/in.}$$

From Equation 7.36, the base-metal shear *rupture* strength per unit length is

$$\phi R_n = 0.45 F_u t = 0.45(58)\left(\frac{9}{16}\right) = 14.7 \text{ kips/in.}$$

The base metal shear strength is therefore 12.2 kips/in. > 10.13 kips/in. (OK)

From Equation 7.29, the weld strength per inch is

$$\phi R_n = \phi(0.707 w F_{nw})$$

The matching electrode for A36 steel is E70. Because the load direction varies on each weld segment, the weld shear strength varies, but for simplicity, we will conservatively use $F_{nw} = 0.6F_{EXX}$ for the entire weld. The required weld size is therefore

$$w = \frac{\phi R_n}{\phi(0.707)F_{nw}} = \frac{10.13}{0.75(0.707)(0.6 \times 70)} = 0.455 \text{ in.}$$

Alternatively, for E70 electrodes, $\phi R_n = 1.392$ kips/in. per sixteenth of an inch in size. The required size in sixteenths is therefore

$$\frac{10.13}{1.392} = 7.3 \text{ sixteenths} \qquad \text{use } \frac{8}{16} \text{ in.} = \frac{1}{2} \text{ in.}$$

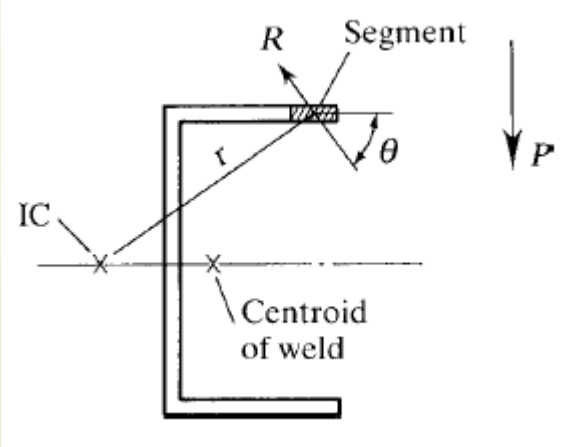
Use a ½-inch fillet weld, E70 electrode.

Maksimum yükle dayanım hesabı

- Kaynaktaki elemanların deformasyonları aşağıdaki formülle hesaplanır

$$\Delta_m = 0.209w(\theta + 2)^{-0.32}$$

Δ_m = deformation of the element at maximum stress
θ = angle that the resisting force makes with the axis of the weld segment (see Figure 8.19)
w = weld leg size



- Maksimum deformasyon $\Delta_u = 1.087w(\theta + 6)^{-0.65} \leq 0.17w$
- Kritik yarıçap kullanılarak diğer elemanlardaki deformasyon

$$\Delta_i = \frac{r_i \Delta_u}{r_{crit}}$$

The stress in each element is then

$$F_{nw} = 0.60F_{EXX}(1 + 0.5\,sin^{1.5}\theta)[p(1.9 - 0.9p)]^{0.3}$$

where

F_{EXX} = weld electrode strength

$p = \frac{\Delta_i}{\Delta_m}$ for the element

Yöntemin çalışma prensibi

1. Solve for the load capacity from the equation

$$\sum M_{IC} = 0$$

   where IC is the instantaneous center.
2. If the two force equilibrium equations are satisfied, the assumed location of the instantaneous center and the load found in Step 1 are correct; otherwise, assume a new location and repeat the entire process.

Örnek

Determine the size of weld required for the bracket connection in Figure 8.18. The service dead load is 10 kips, and the service live load is 30 kips. A36 steel is used for the bracket, and A992 steel is used for the column.

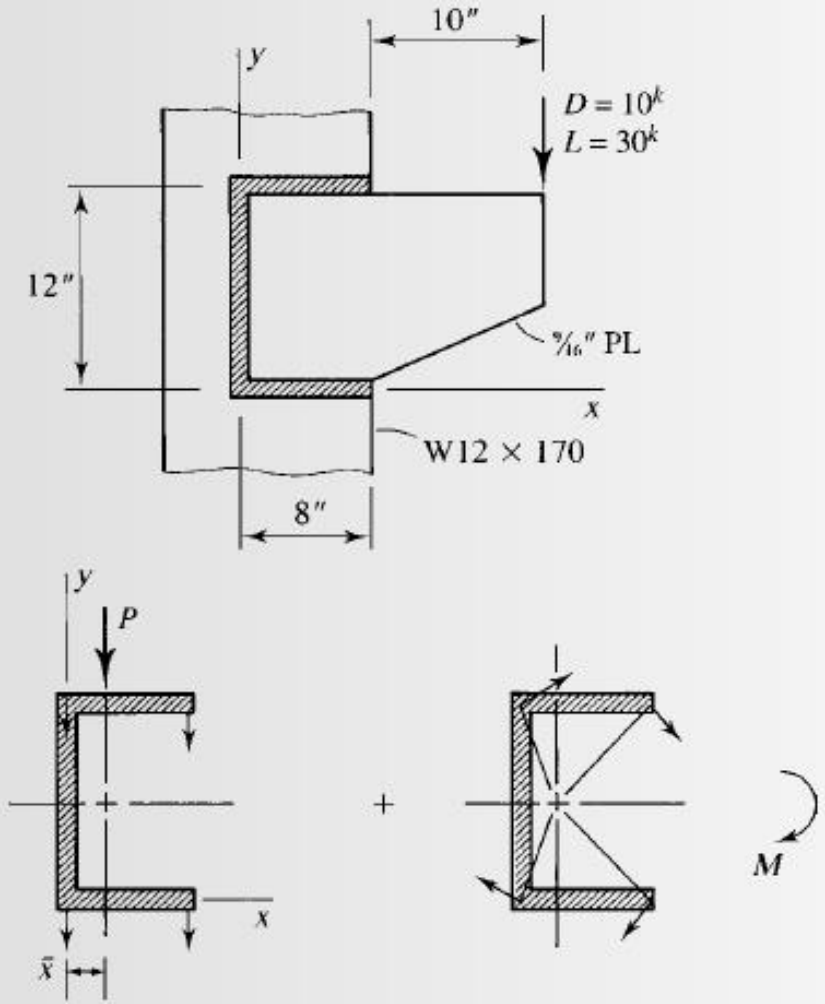


Table 8–8
Coefficients *C*
for Eccentrically Loaded Weld Groups
Angle = 0°

Available Strength of a weld group, ϕR_n or R_n/Ω, is determined with

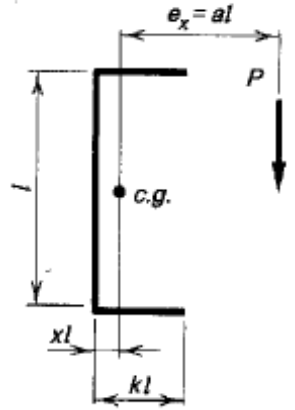
$R_n = CC_1Dl$ ($\phi = 0.75$, $\Omega = 2.00$)

or

LRFD	ASD
$C_{min} = \frac{P_u}{\phi C_1 Dl}$ $D_{min} = \frac{P_u}{\phi C C_1 l}$ $l_{min} = \frac{P_u}{\phi C C_1 D}$	$C_{min} = \frac{\Omega P_a}{C_1 Dl}$ $D_{min} = \frac{\Omega P_a}{C C_1 l}$ $l_{min} = \frac{\Omega P_a}{C C_1 D}$

where

P = required force, P_u or P_a, kips
D = number of sixteenths-of-an-inch in the fillet weld size
l = characteristic length of weld group, in.
$a = e_x / l$
e_x = horizontal component of eccentricity of P with respect to centroid of weld group, in.
C = coefficient tabulated below
C_1 = electrode strength coefficient from Table 8-3 (1.0 for E70XX electrodes)



a	k															
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.2	1.4	1.6	1.8	2.0
0.00	1.86	2.23	2.69	3.25	3.80	4.36	4.92	5.47	6.03	6.59	7.15	8.26	9.37	10.5	11.6	12.7
0.10	1.86	2.28	2.78	3.30	3.83	4.37	4.92	5.46	6.01	6.56	7.11	8.22	9.32	10.4	11.5	12.7

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The weld of Example 8.5 is the same type as the one shown in *Manual* Table 8-8 (angle = 0°), and the loading is similar. The following geometric constants are required for entry into the table:

$$a = \frac{a\ell}{\ell} = \frac{e}{\ell} = \frac{15.7}{12} = 1.3$$

$$k = \frac{k\ell}{\ell} = \frac{8}{12} = 0.67$$

By interpolation in Table 8-8 for $a = 1.3$,

$C = 1.52$ for $k = 0.6$ and $C = 1.73$ for $k = 0.7$

Interpolating between these two values for $k = 0.67$ gives

$C = 1.67$

For E70XX electrodes, $C_1 = 1.0$.

From Table 8-8, the nominal strength of the connection is given by

$$R_n = CC_1D\ell$$

For LRFD,

$$\phi R_n = P_u$$

a	k															
	0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0	1.2	1.4	1.6	1.8	2.0
0.00	1.86	2.23	2.69	3.25	3.80	4.36	4.92	5.47	6.03	6.59	7.15	8.26	9.37	10.5	11.6	12.7
0.10	1.86	2.28	2.78	3.30	3.83	4.37	4.92	5.46	6.01	6.56	7.11	8.22	9.32	10.4	11.5	12.7
0.15	1.83	2.25	2.73	3.23	3.75	4.27	4.80	5.33	5.87	6.41	6.94	8.02	9.11	10.2	11.3	12.4
0.20	1.76	2.18	2.63	3.11	3.60	4.11	4.61	5.13	5.64	6.16	6.68	7.72	8.77	9.83	10.9	12.0
0.25	1.66	2.07	2.51	2.96	3.42	3.90	4.38	4.87	5.37	5.86	6.36	7.37	8.39	9.42	10.5	11.5
0.30	1.55	1.95	2.36	2.79	3.23	3.68	4.14	4.60	5.08	5.55	6.03	7.01	8.00	9.00	10.0	11.0
0.40	1.33	1.69	2.07	2.45	2.84	3.24	3.65	4.07	4.50	4.94	5.39	6.30	7.24	8.19	9.16	10.1
0.50	1.15	1.46	1.79	2.14	2.49	2.85	3.22	3.60	4.00	4.40	4.82	5.67	6.56	7.47	8.40	9.35
0.60	0.999	1.27	1.57	1.88	2.19	2.52	2.85	3.20	3.57	3.94	4.33	5.13	5.97	6.84	7.74	8.65
0.70	0.879	1.12	1.38	1.66	1.95	2.24	2.55	2.87	3.20	3.55	3.91	4.66	5.46	6.29	7.15	8.04
0.80	0.783	0.996	1.23	1.48	1.75	2.02	2.30	2.59	2.90	3.22	3.56	4.27	5.02	5.82	6.64	7.50
0.90	0.704	0.896	1.11	1.34	1.58	1.83	2.09	2.36	2.65	2.95	3.26	3.93	4.65	5.40	6.19	7.01
1.0	0.639	0.813	1.00	1.21	1.44	1.67	1.91	2.16	2.43	2.71	3.01	3.64	4.31	5.03	5.78	6.56
1.2	0.538	0.684	0.845	1.02	1.21	1.42	1.63	1.85	2.08	2.33	2.59	3.15	3.75	4.39	5.07	5.79
1.4	0.464	0.589	0.729	0.883	1.05	1.23	1.42	1.61	1.82	2.04	2.27	2.77	3.31	3.89	4.50	5.15

$$\frac{P_u}{\phi} = CC_1 D\ell$$

and the required value of D is

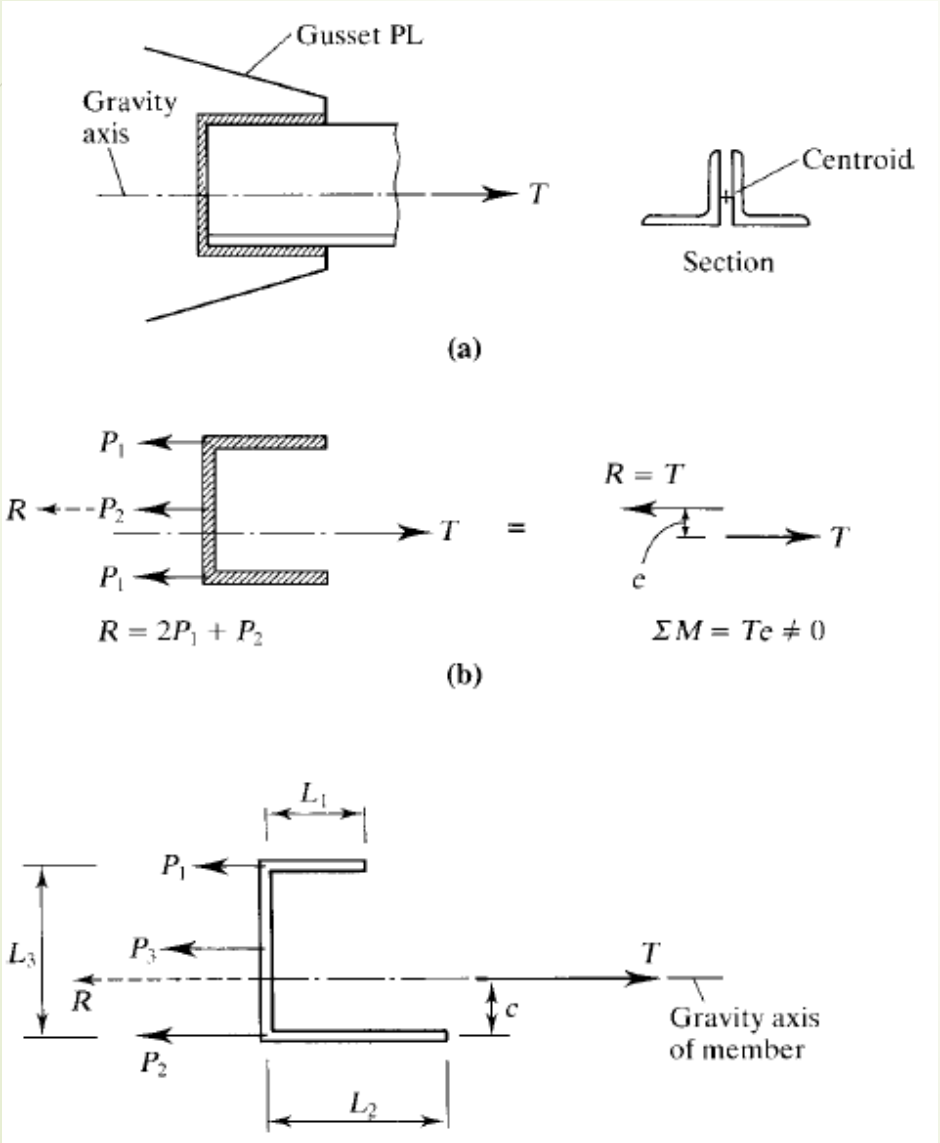
$$D = \frac{P_u}{\phi CC_1 \ell} = \frac{60}{0.75(1.67)(1.0)(12)} = 3.99 \text{ sixteenths}$$

The required weld size is therefore

$$\frac{3.99}{16} = 0.249 \text{ in.} \quad \text{(versus 0.455 inch required in Example 8.5)}$$

Use a ¼-inch fillet weld, E70 electrode.

Çekme durumu için özel durum



$$\sum M_{L_2} = Tc - P_3 \frac{L_3}{2} - P_1 L_3 = 0$$

$$\sum F = T - P_1 - P_2 - P_3 = 0$$

Örnek

A tension member consists of a double-angle section, 2L5 × 3 × ½ LLBB (long legs placed back-to-back). The angles are attached to a ⅜-inch-thick gusset plate. All steel is A36. Design a welded connection, balanced to eliminate eccentricity, that will resist the full tensile capacity of the member.

The nominal strength of the member based on the gross section is

$$P_n = F_y A_g = 36(7.50) = 270.0 \text{ kips}$$

The nominal strength based on the net section requires a value of U, but the length of the connection is not yet known, so U cannot be computed from Equation 3.1. We will use an estimated value of 0.80, and, if necessary, revise the solution after the connection length is known.

$$A_e = A_g U = 7.50(0.80) = 6.000 \text{ in.}^2$$

$$P_n = F_u A_e = 58(6.000) = 348.0 \text{ kips}$$

For A36 steel, the appropriate electrode is E70XX, and

$$\text{Minimum weld size} = \frac{3}{16} \text{ in.} \qquad \text{(AISC Table J2.4)}$$

$$\text{Maximum size} = \frac{1}{2} - \frac{1}{16} = \frac{7}{16} \text{ in.} \qquad \text{(AISC J2.2b)}$$

Compute the required design strength. For yielding of the gross section,

$$\phi_t P_n = 0.90(270.0) = 243.0 \text{ kips}$$

For fracture of the net section,

$$\phi_t P_n = 0.75(348.0) = 261.0 \text{ kips}$$

Yielding of the gross section controls. For one angle, the required design strength is

$$\frac{243.0}{2} = 121.5 \text{ kips}$$

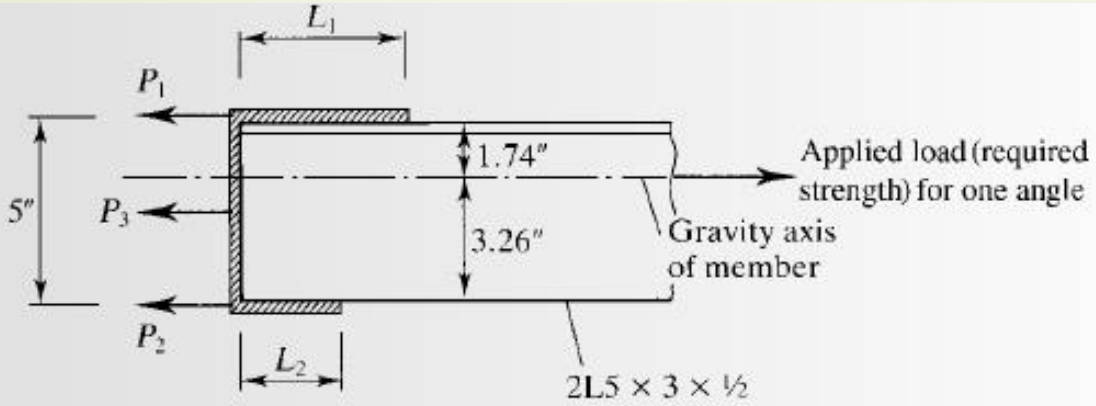
Try a 5/16-inch fillet weld:

Capacity per inch of length = $\phi R_n = 1.392D = 1.392(5) = 6.960$ kips/in.

where D is the weld size in sixteenths of an inch.

Check the base metal shear strength. The gusset plate is the thinner of the connected parts and controls. From Equation 7.35, the base metal shear yield strength per unit length is

$$\phi R_n = 0.6F_y t = 0.6(3\,6)\left(\frac{3}{8}\right) = 8.100 \text{ kips/in.}$$



From Equation 7.36, the base metal shear rupture strength per unit length is

$$\phi R_n = 0.45 F_u t = 0.45(58)\left(\frac{3}{8}\right) = 9.788 \text{ kips/in.}$$

The base metal shear strength is therefore 8.100 kips/in., and the weld strength of 6.960 kips/in. controls.

Refer to Figure 8.22. Since there are both transverse and longitudinal welds, we will try both of the options given in AISC J2.4c. First, we will use the basic electrode strength of $0.6F_{EXX}$ for both the longitudinal and transverse welds (this corresponds to $\phi R_n = 6.960$ kips/in.) The capacity of the weld across the end of the angle is

$$P_3 = 6.960(5) = 34.80 \text{ kips}$$

Summing moments about an axis along the bottom, we get

$$\sum M_{L_2} = 121.5(3.26) - 34.80\left(\frac{5}{2}\right) - P_1(5) = 0, \quad P_1 = 61.82 \text{ kips}$$

$$\sum F = 121.5 - 61.82 - 34.80 - P_2 = 0, \quad P_2 = 24.88 \text{ kips}$$

$$L_1 = \frac{P_1}{6.960} = \frac{61.82}{6.960} = 8.88 \text{ in.} \quad \therefore \text{ Use 9 in.}$$

$$L_2 = \frac{P_2}{6.960} = \frac{24.88}{6.960} = 3.57 \text{ in.} \quad \therefore \text{ Use 4 in.}$$

We will now examine the second option given in AISC J2.4c, in which we use 150% of the basic strength for the transverse weld and 85% of the basic strength for the longitudinal welds. For the transverse weld, use

$$\phi R_n = 1.5 \times 6.960 = 10.44 \text{ kips/in.}$$

and for the longitudinal welds,

$$\phi R_n = 0.85 \times 6.960 = 5.916 \text{ kips/in.}$$

The capacity of the weld across the end of the angle is

$$P_3 = 10.44(5) = 52.20 \text{ kips}$$

Summing moments about an axis along the bottom, we get

$$\sum M_{L_2} = 121.5(3.26) - 52.20\left(\frac{5}{2}\right) - P_1(5) = 0, \quad P_1 = 53.12 \text{ kips}$$

$$\sum F = 121.5 - 53.12 - 52.20 - P_2 = 0, \quad P_2 = 16.18 \text{ kips}$$

$$L_1 = \frac{P_1}{5.916} = \frac{53.12}{5.916} = 8.98 \text{ in.} \quad \therefore \text{ Use 9 in.}$$

$$L_2 = \frac{P_2}{5.916} = \frac{16.18}{5.916} = 2.73 \text{ in.} \quad \therefore \text{ Use 3 in.}$$

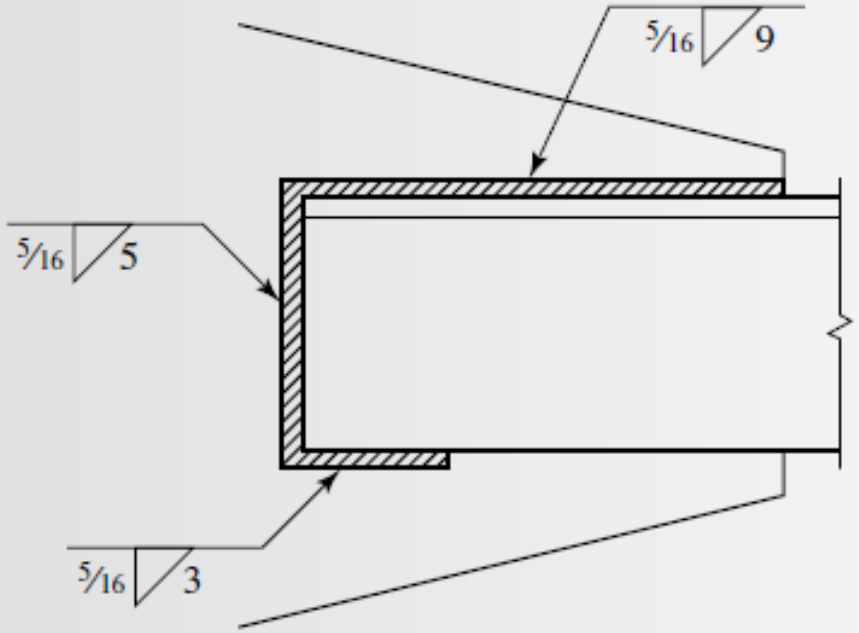
The second option results in a slight savings and will be used.

Verify the assumption on U to be sure that the strength of the member is governed by yielding. From Equation 3.1,

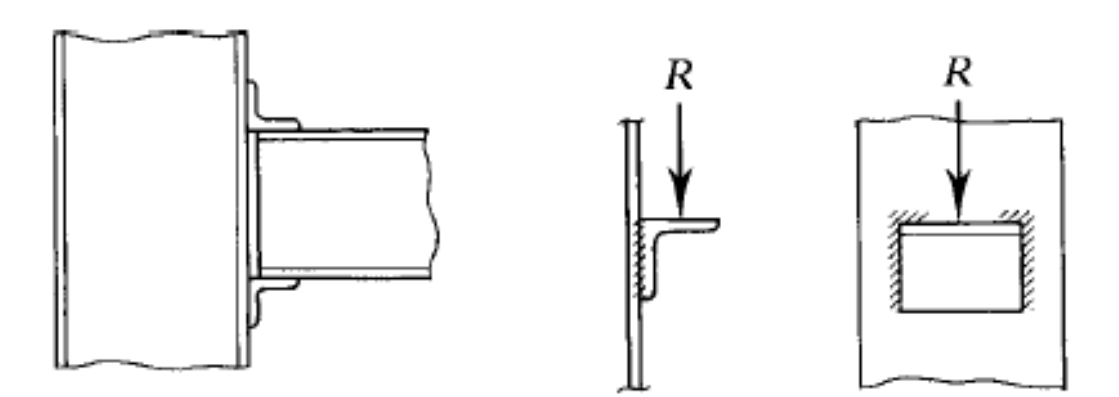
$$U = 1 - \frac{\bar{x}}{\ell} = 1 - \frac{0.746}{9} = 0.9171$$

This is larger than the value of 0.80 that was initially used, so the strength based on fracture will be larger than originally calculated, and the strength will be governed by yielding as assumed.

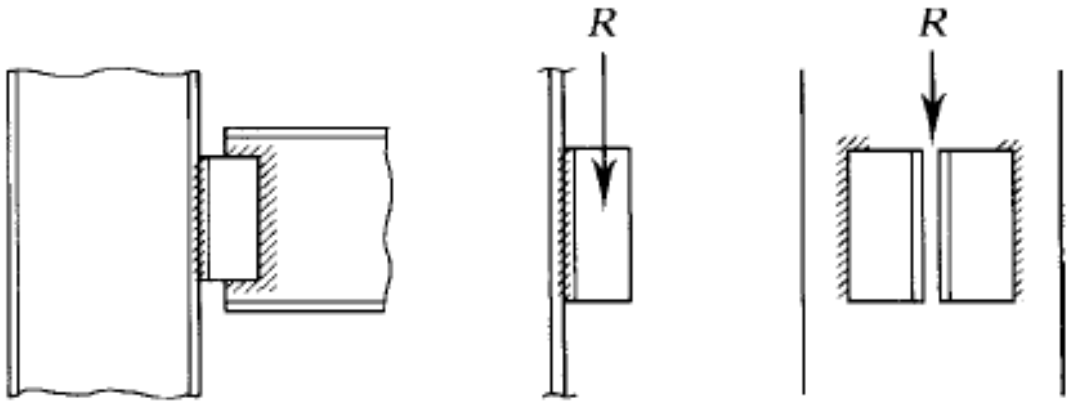
Use the weld shown in Figure 8.23.



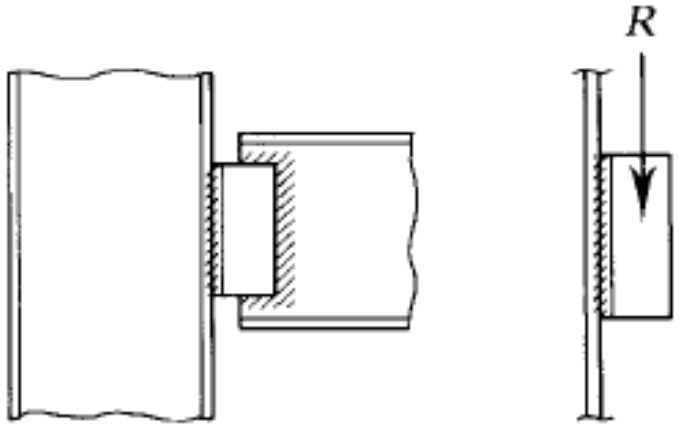
Kesme ve çekme etkisinde kaynak

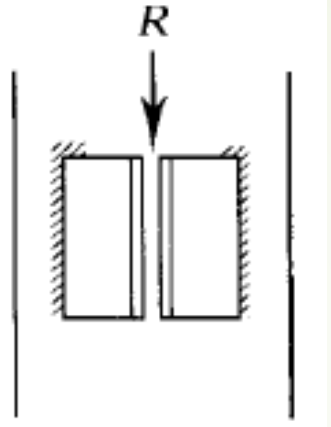


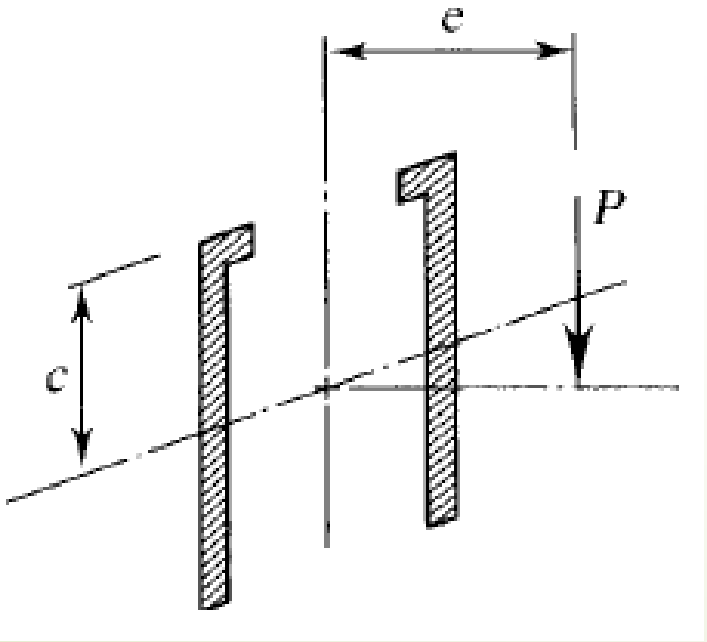
(a) Seated beam connection



(b) Framed beam connection







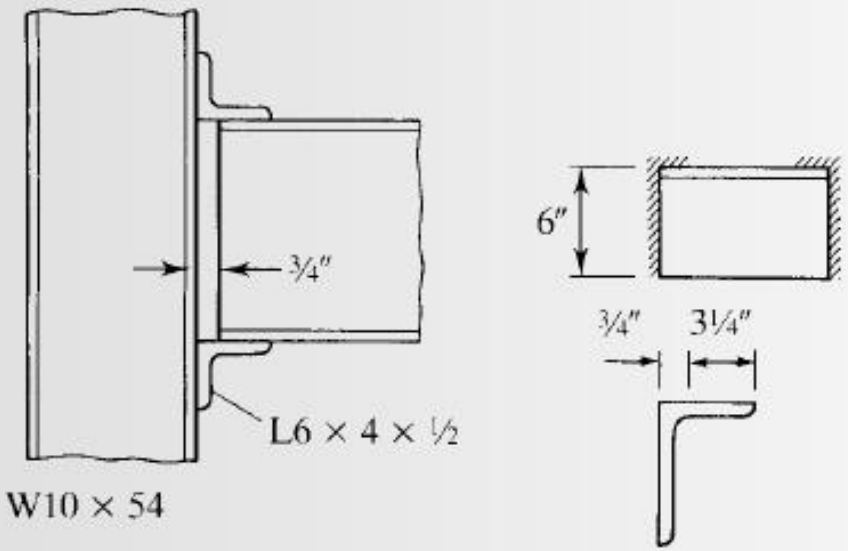
$$f_v = \frac{P}{A}$$

$$f_t = \frac{Mc}{I}$$

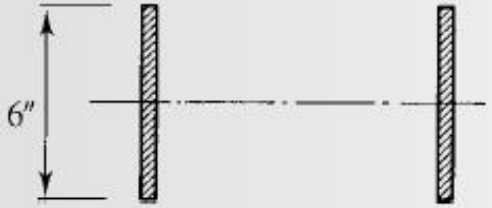
$$f_r = \sqrt{f_v^2 + f_t^2}$$

Örnek

An L6 × 4 × ½ is used in a seated beam connection, as shown in Figure 8.26. It must support a service load reaction of 5 kips dead load and 10 kips live load. The angles are A36 and the column is A992. E70XX electrodes are to be used. What size fillet welds are required for the connection to the column flange?



As in previous design examples, a unit throat size will be used in the calculations. Although an end return is usually used for a weld of this type, conservatively for simplicity it will be neglected in the following calculations.



For the beam setback of ¾ inch, the beam is supported by 3.25 inches of the 4-inch outstanding leg of the angle. If the reaction is assumed to act through the center of this contact length, the eccentricity of the reaction with respect to the weld is

$$e = 0.75 + \frac{3.25}{2} = 2.375 \text{ in.}$$

For the assumed weld configuration in Figure 8.27,

$$I = \frac{2(1)(6)^3}{12} = 36 \text{ in.}^4, \quad c = \frac{6}{2} = 3 \text{ in.}$$

The factored-load reaction is $P_u = 1.2D + 1.6L = 1.2(5) + 1.6(10) = 22$ kips.

$$M_u = P_u e = 22(2.375) = 52.25 \text{ ft-kips}$$

$$f_t = \frac{M_u c}{I} = \frac{52.25(3)}{36} = 4.354 \text{ kips/in.}$$

$$f_v = \frac{P_u}{A} = \frac{22}{2(1)(6)} = 1.833 \text{ kips/in.}$$

$$f_r = \sqrt{f_t^2 + f_v^2} = \sqrt{(4.354)^2 + (1.833)^2} = 4.724 \text{ kips/in.}$$

The required weld size w can be found by equating f_r to the weld capacity per inch of length:

$$f_r = 1.392D$$

$$4.724 = 1.392D, \quad D = 3.394$$

where D is the weld size in sixteenths of an inch for E70 electrodes. The required size is therefore

$$w = \frac{4}{16} = \frac{1}{4} \text{ in.}$$

From AISC Table J2.4,

$$\text{Minimum weld size} = \frac{3}{16} \text{ in.}$$

From AISC J2.2b,

$$\text{Maximum size} = \frac{1}{2} - \frac{1}{16} = \frac{7}{16} \text{ in.}$$

Try $w = \frac{1}{4}$ inch.

Check the shear capacity of the base metal (the angle controls):

Applied direct shear $= f_v = 1.833$ kips/in.

From Equation 7.35, the shear yield strength of the angle leg is

$$\phi R_n = 0.6F_y t = 0.6(36)\left(\frac{1}{2}\right) = 10.8 \text{ kips/in.}$$

From Equation 7.36, the shear rupture strength is

$$\phi R_n = 0.45F_u t = 0.45(58)\left(\frac{1}{2}\right) = 13.1 \text{ kips/in.}$$

The base metal shear strength is therefore 10.8 kips/in. > 1.833 kips/in. (OK)

Use a ¼-inch fillet weld, E70XX electrodes.